

**Topological characterizations of some notable subspaces of the real
line.**

Important applications in General Topology

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Abstract

This work is a summary of the basic topological properties of some important subspaces of the real line: the real line itself, the sets of rational and irrational numbers, the compact interval $[0, 1]$ and the Cantor set, all of them with the subspace topology inherited from $\mathbb{R}$. These spaces were studied by some of the most brilliant topologists of the 20th century, like Brouwer, Sierpinski, Hausdorff or Mazurkiewicz. Moreover, new ways to prove some famous results about these spaces keep appearing to this day. Apart from proving well known topological characterizations of these spaces, we will study their universal properties, that is, their subspaces, their continuous images and their images by quotient maps.

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Introduction

In this thesis we will study some important subspaces of the real line from a topological point of view. Specifically, these spaces are $\mathbb{R}$, $\mathbb{Q}$, the irrationals, which we will denote by $\mathbb{I}$, the Cantor set $\mathcal{C}$ and the interval $[0, 1]$, which we will denote by $\mathbf{I}$. These spaces attracted the interest of mathematicians from the very emergence of set theory and topology as branches of mathematics, between the late nineteenth and early twentieth centuries. As early as 1910, Brouwer proved in [Br10] a surprising topological characterization of $\mathcal{C}$: every metrizable, compact, totally disconnected topological space without isolated points is homeomorphic to $\mathcal{C}$. In doing so, he also showed that it is possible to determine the topology of some spaces by means of a few simple topological properties. Following this idea, Sierpinski proved in [Si20] a characterization of $\mathbb{Q}$, and Hausdorff did the same for $\mathbb{I}$ in [Ha37]. In the first part of the thesis, which corresponds to the first three chapters, these three theorems will be proved.

The characterizations of $\mathcal{C}$ and $\mathbb{I}$ are proved in a very similar way: first we define the spaces $2^{\mathbb{N}}$ and $\mathbb{N}^{\mathbb{N}}$, which are homeomorphic to $\mathcal{C}$ and $\mathbb{I}$ respectively but are easier to work with. We then start with a space X with the properties that characterize $\mathcal{C}$ or $\mathbb{I}$ and construct a homeomorphism between X and $2^{\mathbb{N}}$ or $\mathbb{N}^{\mathbb{N}}$ respectively. This homeomorphism is constructed using an algorithm that is almost the same in both cases.

The characterization of $\mathbb{Q}$ (Sierpinski's Theorem) can also be proved algorithmically by constructing a homeomorphism. For example, this recent article [B119] uses such a method to give an entirely elementary proof of Sierpinski's Theorem. But this undergraduate thesis, drawing on the tools from the previous chapters, will take a different route to prove it. What we will do is prove that $\mathcal{C}$ and $\mathbb{I}$ have a very high degree of homogeneity: they are countable dense homogeneous. This property implies that any pair of countable dense subspaces of $\mathcal{C}$ or $\mathbb{I}$ are homeomorphic, which will allow us to prove Sierpinski's Theorem very easily. The property of countable dense homogeneity is interesting in general, and for this reason we also prove it for $\mathbb{R}$. I also study an even stronger homogeneity property of these spaces, which is included in Appendix A since it moves a little away from the central topic of the thesis.

We will also study universal properties of $\mathcal{C}$, $\mathbb{I}$ and $\mathbb{Q}$: their continuous images, topological subspaces and quotients, and whenever possible we will characterize them up to homeomorphism. Particularly celebrated is the Hausdorff-Alexandroff Theorem, proved in 1927, which states that every compact metric space is the image of the Cantor set under a continuous function (and is therefore a quotient of $\mathcal{C}$).

The second part of the thesis focuses on $\mathbb{R}$ and $\mathbf{I}$. We begin Chapter 4 by proving some properties of order topologies and the usual topology on $\mathbb{R}$. We then move on to study continua, that is, T_2 , compact and connected spaces. These are extensively studied spaces that will help us characterize $\mathbf{I}$ and are a necessary tool for proving the results of the next chapter.

Chapter 5 is devoted to another topic of particular interest to topologists of the late nineteenth and early twentieth centuries: curves, that is, continuous functions from $\mathbf{I}$ to a topological space. In 1890, Peano found something incredible: a continuous surjective function from $\mathbf{I}$ onto $\mathbf{I}^2$, that is, a curve that covered the entire square, and therefore had positive area. The natural question was, what other spaces could be covered by a curve? The well-known Hahn-Mazurkiewicz theorem, first proved by Hahn in 1914, answers that question by stating that a T_2 space can be covered by a curve if and only if it is metrizable, compact, connected and locally connected. A few years later, Mazurkiewicz proves in [Ma20] that every metrizable, locally compact, connected and locally connected space is a continuous image of $\mathbb{R}$, although in this case the converse does not hold. These two theorems are the main objectives of the second part. Another result we prove is that a T_2

space is path connected if and only if it is arc connected, that is, given two points of the space we can find a subspace homeomorphic to $\mathbf{I}$ that contains them. This result may seem deceptively easy to prove, but it is not. Using the results of this chapter, we give a characterization of $\mathbb{R}$ in the last section.

My main objective in writing this thesis has been, as its title indicates, to carry out a systematic study of the topology of the aforementioned subspaces of $\mathbb{R}$ and their universal properties. I have also tried to include many theorems that I find impressive (approximately 1 per section). All of this, of course, within the space constraints of an undergraduate thesis. This means that I have not been able to cover some topics that could have fitted well into this thesis, such as ultrametrics or some embedding theorems, and topics such as continua or descriptive set theory could have been explored further. In any case, it has turned out longer than usual because this is the project I chose to develop upon receiving one of the Departmental Collaboration Scholarships awarded by the Ministry of Education and Vocational Training.

At the beginning of each section I indicate what the content of the section is based on, including references whenever possible. Most of the content of the thesis is taken from the books and articles in the bibliography, although many results are proved differently so that the thesis is self-contained. In particular, most of Chapter 3 and the last two sections of Chapter 5 do not follow any particular source. There are also some ‘original’ results (which I have not found stated anywhere else): Theorem 3.18 and the results of Appendix A.

This thesis is designed to be understandable to a student who has passed a first course in elementary topology. However, there are three well-known results that I use without proof due to lack of space and that may not appear in an introductory topology course:

Theorem 0.1. (*Urysohn’s Metrization Theorem*) *Every regular space with a countable base is metrizable.*

Theorem 0.2. *Every metric space is paracompact.*

Theorem 0.3. (*Classification of 1-manifolds without boundary*) *If X is T_2 , has a countable base, is connected and every point of X has a neighborhood homeomorphic to $\mathbb{R}$, then X is homeomorphic either to $\mathbb{R}$ or to $\mathbb{S}^1$ (the circle).*

Theorem 0.1 is Theorem 23.1 in [Wi70]. A nice one-page proof of 0.2 can be found in [Ru69]. The third, 0.3, is only used in the last section, and is Theorem 5.27 of [Le10].

Chapter 1

The Cantor set, $\mathcal{C}$

In this first chapter we study the Cantor set and a closely related set, $2^{\mathbb{N}}$. We give characterizations of $\mathcal{C}$ as a topological space and as a metric space, characterize which spaces are continuous images of $\mathcal{C}$, and which are homeomorphic to a subspace of it. Finally, we discuss the high degree of homogeneity of $\mathcal{C}$. Apart from the interest of the chapter in its own right, in it we develop tools to prove two of the most important results of this thesis, Sierpinski's theorem and the Hahn-Mazurkiewicz theorem.

1.1 $\mathcal{C}$ and $2^{\mathbb{N}}$

The Cantor set is usually defined geometrically as the set obtained by the following process: We start with the interval $[0, 1]$, which we call C_0 . If we remove its 'middle third', $(\frac{1}{3}, \frac{2}{3})$, we obtain a set C_1 that is the union of two closed intervals, $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. Removing the two middle thirds of these intervals, we obtain a set C_2 consisting of four closed intervals. Through this process we obtain sets C_n that are unions of 2^n closed intervals of length $\frac{1}{3^n}$.

Definition 1.1. The Cantor set is $\mathcal{C} = \bigcap_{n \in \mathbb{N}} C_n$, with its topology as a subspace of $\mathbb{R}$.

It is easy to see that a number x in $[0, 1]$ belongs to C_n iff it can be written in base 3, $x = 0.a_1a_2a_3\dots$ so that the first n digits after the decimal point, $a_1, \dots, a_n$ are 0 or 2. Therefore a number $x \in [0, 1]$ will belong to $\mathcal{C}$ iff x has a base 3 expansion whose digits after the decimal point are all 0 or 2: for example, $\frac{1}{3} = 0.02222\dots$. This alternative definition will be the useful one:

$$\mathcal{C} = \left\{ x = \sum_{n=1}^{\infty} a_n 3^{-n}; a_n \in \{0, 2\} \text{ for each } n \in \mathbb{N} \right\}$$

That is, we can think of the elements of $\mathcal{C}$ as sequences of zeros and twos. The set of sequences of zeros and ones will be important, so let us introduce some notation to be able to discuss it:

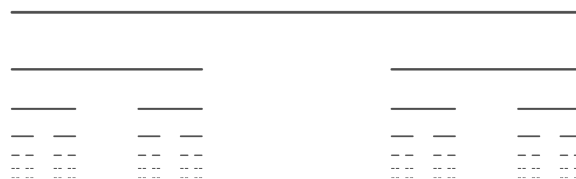


Figure 1.1: The first C_i in the construction of $\mathcal{C}$

Notation 1.2. ($2^{\mathbb{N}}$ and its topology)

We will write $\mathbb{N} = \{1, 2, 3, \dots\}$.

We denote the set of sequences of zeros and ones by $2^{\mathbb{N}} = \{0, 1\}^{\mathbb{N}} = \{(a_n)_{n \in \mathbb{N}}; a_n \in \{0, 1\}\}$.

We denote the set of finite sequences of zeros and ones by $2^{<\mathbb{N}}$, including the sequence of length 0, $()$. Concatenation will be written using the symbol $\wedge$: $(a_1, \dots, a_n) \wedge (b_1, \dots, b_m) = (a_1, \dots, a_n, b_1, \dots, b_m)$.

Finally, 2^n will be the set of sequences of n zeros and ones, and if $(a_n) \in 2^{\mathbb{N}}$, $a|_n$ will be the sequence of the first n terms of a : $a|_n = (a_1, \dots, a_n)$.

Viewing $2^{\mathbb{N}}$ as a countable product of spaces $\{0, 1\}$ with the discrete topology, we can equip it with the product topology. A base of open sets that generates this topology is obtained by associating to each finite sequence of zeros and ones $(b_1, \dots, b_{k-1})$ the open set

$$B_{b_1 \dots b_{k-1}} = \{(a_n) \in 2^{\mathbb{N}}; a_1 = b_1, \dots, a_{k-1} = b_{k-1}\}.$$

That is, $B_{b_1 \dots b_{k-1}}$ contains the sequences of zeros and ones that begin with $b_1, \dots, b_{k-1}$.

Having introduced $2^{\mathbb{N}}$, we will see that it is in fact homeomorphic to the Cantor set. The homeomorphism will map each sequence (a_n) to the number x whose base 3 expansion is $(2a_n)$ (and therefore has a base 3 expansion with only zeros and twos).

Theorem 1.3. *The function*

$$\begin{aligned} f : 2^{\mathbb{N}} &\rightarrow \mathcal{C}; \\ (a_n) &\mapsto \sum_{n=1}^{\infty} 2a_n 3^{-n} \end{aligned}$$

is a homeomorphism.

Proof. Since $2^{\mathbb{N}}$ is compact by Tychonoff's theorem and $\mathcal{C}$ is T_2 (being a subspace of $\mathbb{R}$), it suffices to show that f is bijective and continuous.

The image of f will consist of exactly the numbers that can be written in the form $0.a_1a_2\dots$ in base 3, with a_i being zero or two. This is exactly $\mathcal{C}$, so f is well-defined and surjective.

To show that f is injective, suppose that (a_n) and (b_n) are distinct sequences in $2^{\mathbb{N}}$. Let k be the first position such that $a_k \neq b_k$. Then,

$$f((b_n)) - f((a_n)) = 2 \sum_{i=1}^{\infty} (b_i - a_i) 3^{-i} = 2 \cdot 3^{-k} \left(b_k - a_k + \sum_{i=1}^{\infty} (b_{i+k} - a_{i+k}) 3^{-i} \right).$$

This is not 0, since $|b_k - a_k| = 1$ and $\left| \sum_{i=1}^{\infty} (b_{i+k} - a_{i+k}) 3^{-i} \right| \leq \left| \sum_{i=1}^{\infty} 3^{-i} \right| = \frac{1}{2}$.

Therefore $f((a_n)) \neq f((b_n))$, and f is injective.

It only remains to show that f is continuous. But from what we have just seen, if $(a_n) \neq (b_n)$, then $|f((b_n)) - f((a_n))| \leq 2 \cdot 3^{-k} (1 + \frac{1}{2}) = 3^{-k+1}$, where k is the first digit position with $a_k \neq b_k$. Let us now use the pointwise definition of continuity.

Let $x \in \mathcal{C}$, $x = f((a_n))$ and let $B(x, \varepsilon)$ be a neighborhood of it, with $\varepsilon > 0$. Let k be such that $3^{-k} < \varepsilon$. Then the open neighborhood $B_{a_1 \dots a_k}$ of (a_n) in $2^{\mathbb{N}}$ will satisfy $f(B_{a_1 \dots a_k}) \subseteq B(x, \varepsilon)$ since for any (b_n) in $B_{a_1 \dots a_k}$, (a_n) and (b_n) cannot have different digits before position $k+1$, and therefore $|f((b_n)) - f((a_n))| \leq 3^{-k} < \varepsilon$. $\square$

1.2 Topological characterization of $\mathcal{C}$: Brouwer's Theorem

The content and notation of this section are similar to part of Chapter 10 of [Tr97]. The only addition is Lemma 1.5, since being 0-dimensional and having a countable base are more commonly used properties than having a countable base of clopen sets, so it is preferable for those two properties to be the ones that appear in 1.6.

By 1.3 it is clear that $\mathcal{C}$ is T_1 , compact, has a countable base (being a countable product of spaces with a countable base) and has no isolated points, being a product of infinitely many spaces with more than 1 point. Only one more simple property will be needed to characterize the Cantor set topologically:

Definition 1.4. A topological space is *0-dimensional*¹ if it has a base of clopen sets.

Here I call the open and closed subspaces of a space clopen sets.

Lemma 1.5. *If X is a space with a countable base and is 0-dimensional, then X has a countable base of clopen sets.*

Proof. Let $\{A_n\}_{n \in \mathbb{N}}$ be a countable base of X , and $\mathcal{B}$ a base of clopen sets. Then each A_n also has a countable base, being a subspace of X , so in particular it is Lindelöf. This implies that each A_n has a countable cover by elements $(B_{n,i})_{i \in \mathbb{N}}$ of $\mathcal{B}$ contained in A_n . Then it is clear that the $B_{n,i}$, with $n, i \in \mathbb{N}$, form a countable base of clopen sets of X . $\square$

Theorem 1.6 (Brouwer, 1910). *Let $X \neq \emptyset$ be a topological space that is T_1 , compact, has a countable base, is 0-dimensional and has no isolated points. Then X is homeomorphic to $2^{\mathbb{N}}$.*

The idea of the proof is simple. We can think of the Cantor set as being made by repeatedly dividing the unit interval in 2: as we have seen in its construction, we first divide it in 2 to obtain 2 intervals $I_0 = [0, \frac{1}{3}]$ and $I_1 = [\frac{2}{3}, 1]$.

We then divide these intervals in turn to obtain 4 smaller intervals: $I_{00} = [0, \frac{1}{9}]$, $I_{01} = [\frac{2}{9}, \frac{1}{3}]$, $I_{10} = [\frac{2}{3}, \frac{7}{9}]$, $I_{11} = [\frac{8}{9}, 1]$, and continuing in this way we can define I_ξ for each $\xi \in 2^{<\mathbb{N}}$. The points of the Cantor set are obtained by taking a sequence of increasingly smaller nested intervals and taking their intersection, which will be a single point. We will try to replicate this process in our space X using the properties in the statement.

Proof. By 1.5, X will have a base $(B_n)_{n \in \mathbb{N}}$ of clopen sets. We will now recursively define a clopen set I_ξ of X for each $\xi \in 2^{<\mathbb{N}}$. We define the base case as $I_\emptyset = X$.

Now suppose that we have defined I_ξ and it is nonempty; we will define $I_{\xi \wedge (0)}$ and $I_{\xi \wedge (1)}$ (in the notation of 1.2).

Since I_ξ is open and X has no isolated points, I_ξ has at least 2 points x, y . Since X is T_1 , there is an open set that contains x and not y , so there is some B_k that contains x and not y , that is, $B_k \cap I_\xi$ is a nonempty proper subset of I_ξ . Now let n be the smallest number such that $B_n \cap I_\xi$ is a nonempty proper subset of I_ξ . We define $I_{\xi \wedge (0)} = I_\xi \cap B_n$ and $I_{\xi \wedge (1)} = I_\xi - B_n$. Constructed in this way, the I_ξ will have these properties:

- (1) I_ξ is a nonempty clopen (compact) set for every $\xi \in 2^{<\mathbb{N}}$.
- (2) For every n , $\{I_\xi; \xi \in 2^n\}$ is a partition of X into 2^n clopen sets.
- (3) If ξ has length n and $m < n$, then I_ξ is either contained in B_m or disjoint from B_m .

It is straightforward to prove (1) and (3) by induction on the length of ξ . To prove (2) it suffices to use induction on n and the fact that for any ξ , $I_{\xi \wedge (0)}$ and $I_{\xi \wedge (1)}$ form a partition of I_ξ .

The homeomorphism we want to construct is the following:

$$\begin{aligned} f : 2^{\mathbb{N}} &\rightarrow X; \\ (a_n) &\mapsto y, \text{ where } \{y\} = \bigcap_{n \in \mathbb{N}} I_{a|_n}. \end{aligned}$$

¹There are several definitions of the dimension of a space in topology. In the case of separable metric spaces, which is the case we are concerned with, all notions of a 0-dimensional space coincide with the preceding definition.

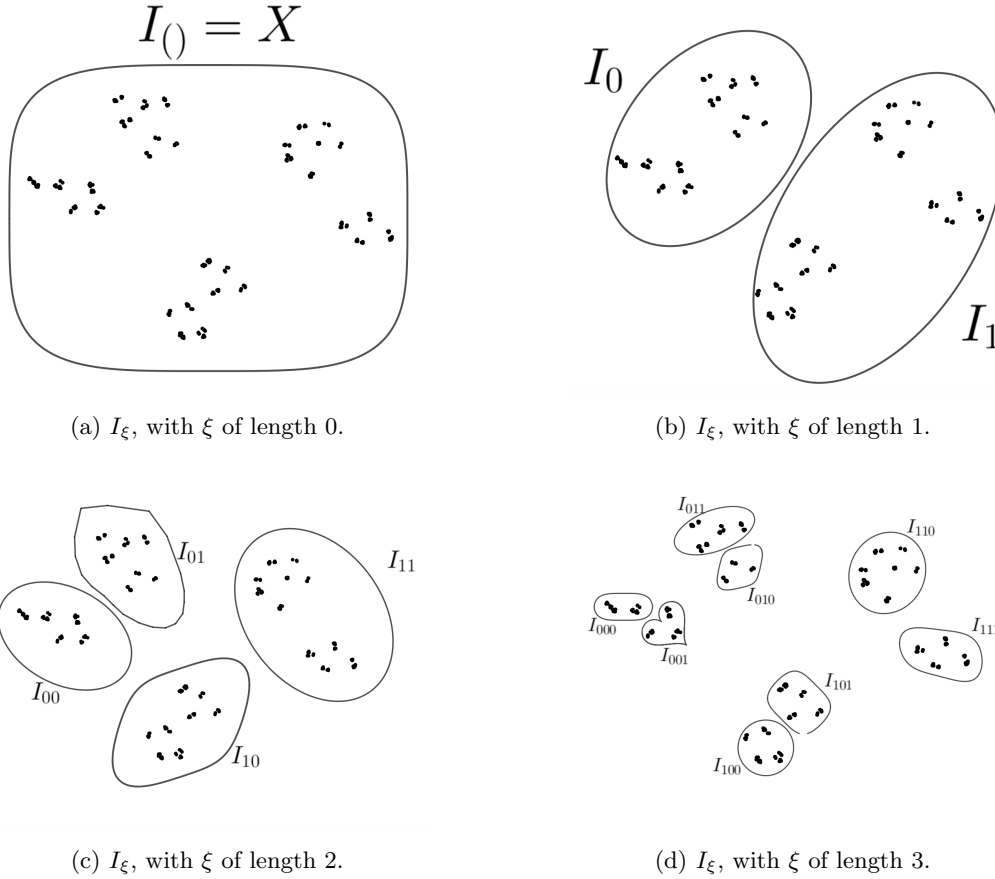


Figure 1.2: The I_ξ of length n , for a given n , are binary partitions of X into 2^n clopen sets.

To show that f is well-defined we need to show that given a sequence (a_n) , $\bigcap_{n \in \mathbb{N}} I_{a|_n}$ has a single point. It is clear that it cannot have more than one point, since if it had two points x, y , there would be some k such that $x \in B_k$ and $y \notin B_k$, and then, since by (3) $I_{a|_{k+1}}$ is either contained in B_k or disjoint from B_k , $I_{a|_{k+1}}$ cannot contain both x and y , contradicting our assumption.

Moreover, $\bigcap_{n \in \mathbb{N}} I_{a|_n}$ cannot be empty since $\{I_{a|_n}\}_{n \in \mathbb{N}}$ has the finite intersection property (it is a chain of nonempty closed sets), and the space X is compact, so the intersection $\bigcap_{n=1}^{\infty} I_{a|_n}$ is nonempty.

Let us now show that f is bijective. It is injective since if $(a_n) \neq (b_n)$, there is some k with $a|_k \neq b|_k$, so by (2) $I_{a|_k} \cap I_{b|_k} = \emptyset$, and hence $f((a_n)) \neq f((b_n))$. It is surjective since given $x \in X$, $x = f((a_n))$, where we can construct (a_n) recursively by choosing a_k at each step so that $x \in I_{a|_k}$.

f is continuous since given a basic open set of X , B_n , by (2) and (3) $f^{-1}(B_n)$ will be the union of some B_ξ , with $\xi \in 2^{n+1}$, and will therefore be open.

f^{-1} is continuous since given a basic open set of $2^{\mathbb{N}}$, B_ξ , $f(B_\xi) = I_\xi$ is a clopen set of X . $\square$

From now on we will say that X is a Cantor set or a Cantor space if it satisfies Theorem 1.6.

Corollary 1.7. *The countable product of finite discrete spaces with ≥ 2 points is a Cantor space.*

Proof. Let $K = \prod_{n \in \mathbb{N}} K_n$, with K_n finite and discrete. K is T_2 since this holds for $K_n \forall n$, it is compact by Tychonoff's theorem, it has a countable base, being a countable product of spaces with a countable base, and it has no isolated points since it is a product of infinitely many spaces with more than one point. Finally, it is 0-dimensional: its subbase as a product space (inverse images of open sets under the projections) consists of clopen sets since the K_n are discrete, so its base consisting of finite intersections of these sets also consists of clopen sets. $\square$

In particular, $n^{\mathbb{N}}$ is homeomorphic to $2^{\mathbb{N}}$ if $n \geq 2$, considering $n = \{0, 1, \dots, n-1\}$ as a discrete space with n points.

1.3 Metric characterization of $\mathcal{C}$

There is a better-known characterization of the Cantor set for metrizable spaces, and in fact it is a consequence of the preceding one. To prove it we will use a couple of results on connectedness in compact T_2 spaces, which will also be useful later in the section on continua. The first, 1.9, is Theorem 6.1.22 of [En89].

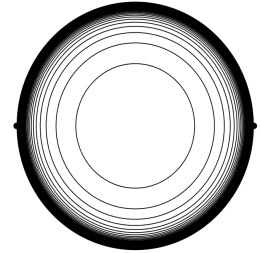
Definitions 1.8. A space X is *connected* if it is not the disjoint union of two nonempty open sets, or equivalently if there is no nonempty proper clopen subspace of X .

A *connected component* of a space is a connected subspace that is maximal with respect to inclusion.

The *quasicomponent* of a point x in a space X is the intersection of all clopen sets in X that contain x .

A space X is *totally disconnected* if for every $x \in X$, $\{x\}$ is a connected component, that is, if there are no connected subspaces with more than 1 point.

We know that components and quasicomponents are closed, and that they form partitions of the space. Moreover, it is easy to see that the connected component of a point is contained in its quasicomponent, using the fact that, being connected, the component cannot have any proper clopen subset. However, the converse is not true in general. For example, if we consider the set in the figure consisting of the points $(-1, 0)$, $(1, 0)$ and the circles with center 0 and radius $1 - \frac{1}{n}$ for $n \geq 2$, the component of the point $(1, 0)$ contains only that point. However, it is not difficult to see that the quasicomponent of $(1, 0)$ is the set $\{(1, 0), (-1, 0)\}$. Nevertheless, in the case of compact T_2 spaces the reverse inclusion does hold:



Lemma 1.9. *If X is a compact T_2 space and $x \in X$, the component of x and its quasicomponent coincide.*

Proof. Since the component is contained in the quasicomponent, it will suffice to check that the quasicomponent of x , which we will call Q , is connected. Q is closed, being the intersection of the clopen sets that contain x . Suppose that there is a separation of Q , $Q = X_1 \cup X_2$, with X_1 and X_2 disjoint and closed in Q and $x \in X_1$. Then X_1 and X_2 are closed in X , so since X is normal, there are disjoint open sets U, V in X with $X_1 \subseteq U$, $X_2 \subseteq V$. $U \cup V$ is a neighborhood of Q . If we denote by $\{F_\alpha\}_\alpha$ the clopen sets that contain x , then $\{X - F_\alpha\}_\alpha$ is an open cover of the compact set $X - (U \cup V)$. Therefore there will be finitely many $F_{\alpha_1}, \dots, F_{\alpha_n}$ whose complements cover $X - (U \cup V)$. Therefore, $F = \bigcap_{i=1}^n F_{\alpha_i}$ is a clopen neighborhood of Q contained in $U \cup V$.

Then we know that $U \cap F$ is open, and moreover, $\overline{U \cap F} \subseteq \overline{U} \cap \overline{F} = \overline{U} \cap F = \overline{U} \cap (U \cup V) \cap F = U \cap F$. Therefore $U \cap F$ is a clopen set, and it contains x , so $Q \subseteq U \cap F$. But then $X_2 \subseteq Q \cap V \subseteq U \cap F \cap V = \emptyset$, that is, $X_2 = \emptyset$. This proves that Q is connected. $\square$

Lemma 1.10. *Let X be a compact, T_2 and totally disconnected space. Then X is 0-dimensional.*

Proof. Let $x \in X$ and let U be a neighborhood of x . By 1.9, since $\{x\}$ is the component of x , for each $y \neq x$ there will be a clopen set that contains y but not x , O_y . Since the O_y cover $X - U$, which is compact, there will be finitely many clopen sets $O_{y_1}, \dots, O_{y_n}$ that cover $X - U$. Writing $C = \bigcap_{i=1}^n (X - O_{y_i})$, C is a clopen set with $x \in C \subseteq U$, as desired. $\square$

We are now ready to prove the characterization.

Theorem 1.11. *Let $X \neq \emptyset$ be a metrizable, compact, totally disconnected space without isolated points. Then X is a Cantor set.*

Proof. To use Theorem 1.6 we need to show that X is 0-dimensional and has a countable base. By 1.10, X will be 0-dimensional. It will also have a countable base because, being compact, it is Lindelöf, and the properties of being Lindelöf and having a countable base are equivalent in metric spaces. $\square$

1.4 Continuous images of $\mathcal{C}$. The Hausdorff-Alexandroff Theorem

Let us see which spaces are continuous images of the Cantor set, that is, for which spaces X there exists a continuous surjective function $f : \mathcal{C} \rightarrow X$. Obviously, such spaces X will have to be compact. For convenience we will ignore spaces that are not T_2 , since in that case, as can be seen in [DS14], pathological spaces arise. In 1.14 we will obtain a nice characterization of the T_2 continuous images of the Cantor set. First we prove, as in [Sc74], the well-known Hausdorff-Alexandroff Theorem, which tells us that every compact metric space is a continuous image of $\mathcal{C}$.

Theorem 1.12 (Hausdorff-Alexandroff, 1927). *Let $X \neq \emptyset$ be a compact metric space. Then there is a continuous surjective function $f : \mathcal{C} \rightarrow X$.*

In the proof we will use the fact that every compact metric space X is totally bounded: given $\varepsilon > 0$, X (and therefore any subset of it) has a cover by finitely many subsets of diameter $< \varepsilon$. In fact, also for later use, we will prove a lemma that characterizes compact metric spaces.

Lemma 1.13. *A metric space (X, d) is compact if and only if it is complete (2.3) and totally bounded.*

Proof. Suppose that X is compact. Given $\varepsilon > 0$, $\{B(x, \frac{\varepsilon}{3}); x \in X\}$ is a cover of X by open sets of diameter $< \varepsilon$. Taking a finite subcover shows that X is totally bounded. If (x_n) is a Cauchy sequence in X , it will have an accumulation point x . Given $\varepsilon > 0$, we choose N such that $\forall m, n \geq N, d(x_m, x_n) < \frac{\varepsilon}{2}$ and $d(x_N, x) < \frac{\varepsilon}{2}$. Then for every $n > N, d(x_n, x) < \varepsilon$. This shows that x_n is convergent, and X is complete.

Now suppose that X is complete and totally bounded; let us show that every sequence x_n has an accumulation point. Let $\mathcal{B}_n$ be a finite cover of X by subsets of diameter $< \frac{1}{n}$. There is a member of $\mathcal{B}_1$ that contains infinitely many terms of the sequence, and therefore there is a subsequence $x_{1,n}$ of x_n that has diameter < 1 . Using this process recursively, we construct for each $i \in \mathbb{N}$ a sequence $(x_{i,n})_{n \in \mathbb{N}}$ that will be a subsequence of $x_{i-1,n}$ contained in an element of $\mathcal{B}_i$, and therefore have diameter $< \frac{1}{i}$. If we now take the subsequence $x_{n,n}$ of x_n , it is Cauchy, and therefore converges to some $x \in X$. Therefore x_n has an accumulation point. $\square$

The idea of the proof of Theorem 1.12 will be to construct a series of covers $\mathcal{U}_n$ of X by subsets of diameter $\leq \frac{1}{n}$, so that $\mathcal{U}_{n+1}$ will be a refinement of $\mathcal{U}_n$ for each n . Using these covers, we will find a Cantor set K and a fairly natural continuous map from K onto X .

Proof of Theorem 1.12. Let X be our space. It will have a cover by finitely many nonempty subsets of diameter ≤ 1 , which we will call $\mathcal{U}_1 = \{U_i\}_{i=1, \dots, k_1}$. For technical reasons (see 1.7), we set $k_1 > 1$.

Now, for $k_2 > 1$ sufficiently large, we will be able to cover each U_i , from $i = 1$ to k_1 , with k_2 nonempty subsets of it of diameter $\leq \frac{1}{2}$. We thus form a finite collection $\mathcal{U}_2 = \{U_{i_1 i_2}\}_{i_1=1, \dots, k_1, i_2=1, \dots, k_2}$, so that $\bigcup_{i_2=1, \dots, k_2}^{k_2} U_{i_1 i_2} = U_{i_1}$ for every i . Thus, in particular, $\mathcal{U}_2$ also covers X .

That is, intuitively, $\mathcal{U}_2$ consists of covering each element of $\mathcal{U}_1$ with k_2 subsets of diameter $< \frac{1}{2}$. We will continue this process by constructing collections $\mathcal{U}_n$ so that each element of $\mathcal{U}_{n-1}$ is covered by k_n subsets from $\mathcal{U}_n$ of diameter $< \frac{1}{n}$. We construct the $\mathcal{U}_n$ recursively:

For each $n \in \mathbb{N}$ there is some $k_n > 1$ large enough that we can construct a finite collection of nonempty subsets of diameter $\leq \frac{1}{n}$, $\mathcal{U}_n = \{U_{i_1 \dots i_n}\}_{i_j=1, \dots, k_j}$, so that for all $i_1, \dots, i_{n-1}$,

$$U_{i_1, \dots, i_{n-1}} = \bigcup_{j=1}^{k_n} U_{i_1, \dots, i_{n-1}, j}.$$

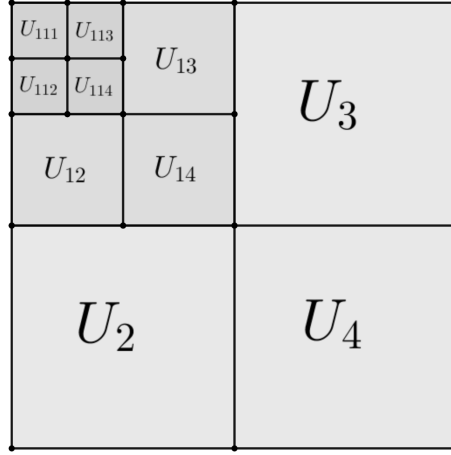


Figure 1.3: Example of the construction of the $\mathcal{U}_n$ when X is a square. In this case, $\mathcal{U}_n$ is obtained by dividing each square of $\mathcal{U}_{n-1}$ into 4 squares. In particular, $k_n = 4 \forall n$.

We have now constructed the covers of X . Now, writing K_n for the discrete space $\{1, \dots, k_n\}$, we consider the topological product:

$$K = \prod_{n=1}^{\infty} K_n, \text{ the set of sequences } s = (s_n)_{n \in \mathbb{N}} \text{ with } s_n \in \{1, \dots, k_n\}.$$

By Corollary 1.7, K is a Cantor space. We will try to find a continuous function F from K onto X . To do so, given a sequence $s = (s_n) \in K$, we consider the decreasing sequence of nonempty compact subsets $A_{s,n} = \overline{U_{s_1 \dots s_n}}$. Their intersection will be nonempty by Cantor's intersection theorem, and moreover, since their diameters tend to 0, $\bigcap_{n \in \mathbb{N}} A_{s,n}$ will have exactly one point.

Thus each $s \in K$ will have an associated point $x_s = \bigcap_{n \in \mathbb{N}} A_{s,n}$.

As one would expect from the way things have developed, the continuous function we are looking for will be:

$$\begin{aligned} F : K &\rightarrow X; \\ s = (s_n)_{n \in \mathbb{N}} &\mapsto x_s \end{aligned}$$

Let us show that F is continuous at each point: let s be a sequence, and consider the ball $B(x_s, \frac{1}{n})$ in X . The set $B_{s|_{n+1}}$ of sequences r with $r_i = s_i$ for $i = 1, \dots, n+1$ is an open set of K , so it will suffice to show that $f(B_{s|_{n+1}}) \subseteq B(x_s, \frac{1}{n})$ to show that f is continuous.

Given $r \in B_{s|_{n+1}}$, we have $A_{s,n+1} = A_{r,n+1}$. Therefore, since $A_{s,n+1}$ has diameter $\leq \frac{1}{n+1}$ and $x_r, x_s \in A_{s,n+1}$, we will have $d(F(s), F(r)) = d(x_s, x_r) < \frac{1}{n}$. Thus F is continuous.

Finally, we check that F is surjective. Given $x \in X$, there is some $s_1 \in \{1, \dots, k_1\}$ such that $x \in U_{s_1}$. Since $U_{s_1 j}$, $j = 1, \dots, k_2$ cover U_{s_1} , there is some s_2 such that $x \in U_{s_1 s_2}$. Repeating this reasoning we obtain a sequence $s = (s_n)$ such that $x \in U_{s_1 \dots s_n}$ for every n . Therefore $x = F(s)$, since $x \in A_{s,n} \forall n$. $\square$

Thus we now know that all compact metric spaces are continuous images of Cantor space. In fact, the following result states that this characterizes all compact T_2 spaces that are continuous images of the Cantor set:

Theorem 1.14. *Let $X \neq \emptyset$ be a compact T_2 space. The following are equivalent:*

1. X is a continuous image of the Cantor set
2. X has a countable base

3. X is metrizable

4. X is a quotient of the Cantor set

To prove it we will use this lemma (Corollary 23.2 in [Wi70]):

Lemma 1.15. *If X is compact and metric, Y is T_2 and $f : X \rightarrow Y$ is continuous and surjective, then Y is metrizable.*

Proof. We know that Y is T_2 , and compact, being a continuous image of X , so it is regular. Thus by Urysohn's theorem 0.1 it suffices to show that Y has a countable base.

Since X is metrizable and compact, it will be metric and Lindelöf, and will therefore have a countable base. Let $\mathcal{B}$ be a countable base of X and let $\mathcal{D}$ be the set of finite unions of members of $\mathcal{B}$, which is also countable. Since the function f is closed, $Y - f(X - D)$ is open for every $D \in \mathcal{D}$.

We want to show that the countable collection $\mathcal{E} := \{Y - f(X - D); D \in \mathcal{D}\}$ is a base for the topology of Y . Let $p \in Y$ and let U be an open neighborhood of p . Then $f^{-1}(p) \subseteq f^{-1}(U)$, and $f^{-1}(p)$ is compact, being closed in X . Therefore we can choose finitely many open sets $B_1, \dots, B_n$ from $\mathcal{B}$ such that $f^{-1}(p) \subseteq B_1 \cup \dots \cup B_n \subseteq f^{-1}(U)$. Therefore, taking $D = B_1 \cup \dots \cup B_n \in \mathcal{D}$, we have $p \in Y - f(X - D) \subseteq U$, as desired. $\square$

Proof of Theorem 1.14.

- 1 $\iff$ 4 Every quotient of the Cantor set is a continuous image of it (under the quotient map). Moreover, if $f : \mathcal{C} \rightarrow X$ is continuous and surjective, since $\mathcal{C}$ is compact and X is T_2 , it will be closed and therefore a quotient map.
- 2 $\iff$ 3 We already know that every compact metric space has a countable base, and if a compact T_2 space has a countable base then it is regular and has a countable base, so by 0.1 it is metrizable.
- 1 $\iff$ 3 By Lemma 1.15 and the Hausdorff-Alexandroff theorem 1.12.

$\square$

1.5 Subspaces of $\mathcal{C}$. Another characterization of $\mathcal{C}$

The following characterization of the subspaces of $\mathcal{C}$ can be found in Chapter 10 of [Tr97].

Theorem 1.16. *A space X is homeomorphic to a subspace of $\mathcal{C}$ if and only if it is T_0 , has a countable base and is 0-dimensional.*

Proof. One implication is obvious; let us prove the other. If X is T_0 , has a countable base and is 0-dimensional, it has a countable base of clopen sets, $\{C_n\}_{n \in \mathbb{N}}$. We consider the following function:

$$f : X \rightarrow 2^{\mathbb{N}};$$

$$x \mapsto (a_n) \quad , \text{ with } a_n = \begin{cases} 0 & \text{if } x \notin C_n \\ 1 & \text{if } x \in C_n \end{cases}$$

Then f is injective since X is 0-dimensional. It is also continuous, since the inverse images of open sets from the subbase of $2^{\mathbb{N}}$ are the clopen sets C_n and their complements. Moreover, $f(C_n)$ will consist of the elements (a_n) of $f(X)$ that satisfy $a_n = 1$, which form an open set in $f(X)$. Therefore f is a homeomorphism from X to $f(X)$. $\square$

If we turn our attention to the open subsets of $\mathcal{C}$, we will see that, surprisingly, there are only 2 classes of open subspaces up to homeomorphism: the compact ones and the noncompact ones. To prove this we use a lemma.

Lemma 1.17. *Every open subspace $\neq \emptyset$ of the Cantor set is a finite or countably infinite union of disjoint clopen sets.*

Proof. If $A \subseteq \mathcal{C}$ is open, since $\mathcal{C}$ has a countable base $\mathcal{B}$ of clopen sets, we can take those that are contained in A , so that A will be a countable union of clopen sets, $A = \bigcup_{n \in \mathbb{N}} B_n$. Now taking $C_n = B_n - \bigcup_{i=1}^{n-1} B_i$, we have that the C_n are disjoint clopen sets whose union is A . $\square$

Theorem 1.18. *Let A be an open subspace of $\mathcal{C}$. If A is compact (closed), then it is a Cantor set. Otherwise, A is homeomorphic to $\mathcal{C} - \{0\}$.*

Proof. The first part follows directly from 1.11, since, being open, A will have no isolated points.

If A is not compact, by 1.17 it will be a countable disjoint union of clopen sets. The union cannot be finite since then A would be compact. We will prove the second part by showing that any two noncompact open subsets of $\mathcal{C}$ are homeomorphic.

If A, B are noncompact open subsets of $\mathcal{C}$, then $A = \bigcup_{n \in \mathbb{N}} A_n$ and $B = \bigcup_{n \in \mathbb{N}} B_n$, with (A_n) and (B_n) sequences of disjoint nonempty clopen sets. By the first part of the theorem, A_n and B_n are all Cantor sets, and there will be a homeomorphism $f_n : A_n \rightarrow B_n$.

Combining all these homeomorphisms, we obtain a bijection $f : A \rightarrow B$ with $f|_{A_n} = f_n$. This function will in fact be a homeomorphism. To see this, by symmetry it suffices to show that it is continuous. Given an open set U of B , $U \cap B_n$ will be open in B_n , so $f^{-1}(U \cap B_n) = f_n^{-1}(U \cap B_n)$ will be open in A_n , and will therefore be open in A . Thus $f^{-1}(U) = \bigcup_n f^{-1}(U \cap B_n)$ is open in A . $\square$

Thus both $\mathcal{C}$ and $\mathcal{C} - \{0\}$ have exactly one class of (nonempty) compact open subsets and exactly one class of noncompact open subsets. This property is very restrictive, and we will see that, in fact, $\mathcal{C}$ and $\mathcal{C} - \{0\}$ are the only metric spaces that satisfy it. To prove this characterization, which can be found in [SG75], we will need a tool that we will not develop until the beginning of the chapter on continua. That chapter is independent of the rest, so this will not be a problem. The result we use is 4.11, which, expressed in elementary terms, says:

Given a space X that is T_2 , compact and connected with ≥ 2 points, there is some $x \in X$ such that $X - \{x\}$ is connected.

Theorem 1.19. *(Characterization of $\mathcal{C}$ and of $\mathcal{C} - \{0\}$) Let X be a metric space that satisfies these two properties:*

- (1) *X has nonempty compact open subspaces, and they are all homeomorphic.*
- (2) *X has noncompact open subspaces, and they are all homeomorphic.*

Then, if X is compact it is homeomorphic to $\mathcal{C}$, and if X is not compact it is homeomorphic to $\mathcal{C} - \{0\}$.

Proof. Suppose that X satisfies the statement and is compact. By 1.11, it suffices to check that X is totally disconnected and has no isolated points. If X had isolated points, by (1) they would be homeomorphic to X , and then, since X would have a single point, X would not satisfy (2). Therefore X will have no isolated points (in particular, it is infinite). Let us now show that X is totally disconnected.

Let x, y be two points of X at distance d . Then the union of balls $U = B(x, \frac{d}{3}) \cup B(y, \frac{d}{3})$ is clearly disconnected. If U is compact, and therefore homeomorphic to X , then X is disconnected. Suppose instead that U is not compact. Since every $p \in X$ is nonisolated, $X - \{p\}$ is not compact. Therefore, by (2), $X - \{p\}$ will be homeomorphic to U , hence disconnected. Since $X - \{p\}$ is disconnected $\forall p$, X will be disconnected by the result stated before the theorem.

In both cases, we know that X is disconnected. Now, given a point of X , let C be its connected component. Let us show that it has only one point. C is closed. It cannot be open since then by (1) it would be disconnected. Therefore $X - C$ is not compact, hence it is homeomorphic to $X - \{x\}$. Now, by Lemma 1.9, C is the quasicomponent of x in X . Therefore for each y in $X - C$, there is a clopen neighborhood of y disjoint from C . That is, each y in $X - C$ lies in a compact open subset of $X - C$. Since $X - C$ is homeomorphic

to $X - \{x\}$, any y in $X - \{x\}$ lies in a compact open subset of $X - \{x\}$. This implies that X is totally disconnected, since any pair of points is separated by clopen sets. Therefore, by 1.11, the case where X is compact is proved.

Now suppose that X is not compact. By (1), it will have a compact open subset. This subset will also satisfy properties (1) and (2), so it will be homeomorphic to $\mathcal{C}$. Thus X has an open subset homeomorphic to $\mathcal{C} - \{0\}$, and by (2) we are done. $\square$

1.6 Homogeneity of $\mathcal{C}$. Countable dense homogeneity

Definition 1.20. We say that a space X is homogeneous if, given two of its points x, y , there is a homeomorphism $f : X \rightarrow X$ with $f(x) = y$.

This means that any two points of the space are topologically indistinguishable. A broad class of homogeneous spaces is the following:

Definition 1.21. A topological group is a triple $(X, *, \mathcal{T})$, where X is a set, $(X, *)$ is a group, and $\mathcal{T}$ is a topology on X such that the functions $*$: $X \times X \rightarrow X; (x, y) \mapsto x * y$ and $\bullet^{-1} : X \rightarrow X; x \mapsto x^{-1}$ are continuous.

Example 1.22. $\mathbb{R}$ and $\mathbb{Q}$ are topological groups under addition.

$2^{\mathbb{N}}$ is a topological group, defining addition as $(a_n) + (b_n) = (a_n + b_n)$, where $a_n + b_n$ is addition in $\mathbb{Z}/2\mathbb{Z}$, that is, with $1+1=0$. The function $\bullet^{-1}$ is the identity in this case, and it is straightforward to see that the function $+$ is continuous.

In a topological group, the function $\bullet * y : x \mapsto x * y$ is continuous, since it is obtained by composing the two continuous functions $x \mapsto (x, y) \mapsto x * y$. Thus, given two points x, y , the functions $\bullet * x^{-1} * y$ and $\bullet * y^{-1} * x$ are continuous and inverse to each other, so they are homeomorphisms that interchange x and y . This proves the following proposition:

Proposition 1.23. *Topological groups are homogeneous.* $\square$

Thus, in particular, $\mathbb{R}, \mathbb{Q}$ and $\mathcal{C}$ will all be homogeneous.

Just as homogeneity tells us when any two points of a set are indistinguishable, we can consider when any two finite (ordered) subsets of a set are indistinguishable:

Definition 1.24. We say that a space X is strongly n -homogeneous when, given distinct $x_1, \dots, x_n \in X$ and distinct $y_1, \dots, y_n \in X$, there is a homeomorphism $f : X \rightarrow X$ with $f(x_i) = y_i$, for $i = 1, \dots, n$.

For example, it is easy to see that $\mathbb{R}$ is strongly 2-homogeneous: given $x_1, x_2, y_1, y_2 \in \mathbb{R}$, the homeomorphism

$f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto y_1 + (x - x_1) \frac{y_2 - y_1}{x_2 - x_1}$ satisfies $f(x_i) = y_i$. However, $\mathbb{R}$ is not strongly 3-homogeneous.

This is because a homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$ is always monotone, so it could not satisfy $f(0) = 1, f(1) = 2, f(2) = 0$. A slightly surprising result is that any connected manifold of dimension ≥ 2 is strongly n -homogeneous for every n . The same holds for the Cantor set:

Proposition 1.25. *$\mathcal{C}$ is strongly n -homogeneous for every n .*

Proof. Let $x_1, \dots, x_n, y_1, \dots, y_n \in \mathcal{C}$. We can find clopen sets $A_1, \dots, A_n, B_1, \dots, B_n$ that are neighborhoods of $x_1, \dots, x_n, y_1, \dots, y_n \in \mathcal{C}$ respectively, so that $A_1, \dots, A_n$ are disjoint, $B_1, \dots, B_n$ are disjoint, and $A_0 = X - \bigcup_{i=1}^n A_i$ and $B_0 = X - \bigcup_{i=1}^n B_i$ are nonempty.

Then, by 1.18, $A_0, A_1, \dots, A_n, B_0, B_1, \dots, B_n$ are all Cantor sets, so we have homeomorphisms $f_0 : A_0 \rightarrow B_0, f_1 : A_1 \rightarrow B_1, \dots, f_n : A_n \rightarrow B_n$. Since $A_1, \dots, A_n$ are homogeneous, these homeomorphisms f_i can be chosen so that $f_i(x_i) = y_i$ for $i = 1, \dots, n$. The function $f : \mathcal{C} \rightarrow \mathcal{C}$ defined by $f_0, f_1, \dots, f_n$ will be a homeomorphism by the pasting lemma. $\square$

A type of homogeneity that, a priori, seems more restrictive is countable dense homogeneity:

Definition 1.26. Given a separable space X , we say that X is countable dense homogeneous if, given two countable dense subspaces D, E of X , there is a homeomorphism $f : X \rightarrow X$ with $f(D) = E$.

This property means not only that any pair of countable dense subspaces are homeomorphic, but that this homeomorphism can be extended to the whole space. The reason we have mentioned this property in this chapter is, naturally, that $\mathcal{C}$ is countable dense homogeneous. This result can be found as part *e* of Exercise 4.3H in [En89]. However, we can give a proof that is essentially the same as that of 1.6. To explain the idea of the proof, which is based on [Eb77], we will need some notation for $2^{\mathbb{N}}$:

Notation 1.27. Given $\xi = (\xi_1, \dots, \xi_k) \in 2^{<\mathbb{N}}$, we denote by $\bar{\xi} \in 2^{\mathbb{N}}$ the result of completing ξ with zeros, $(\xi_1, \dots, \xi_k, 0, 0, \dots)$.

We will denote by $\overline{2^{<\mathbb{N}}}$ the subset of $2^{\mathbb{N}}$ consisting of its elements with finitely many nonzero terms, that is, the sequences that are 0 from some point onwards.

In the proof of 1.6 we showed that if X is T_1 , compact, has a countable base, is 0-dimensional and has no isolated points, then it is homeomorphic to $\mathcal{C}$. To do this we formed successive partitions of X into 2^n clopen subsets of it (the I_ξ), and used the partitions to construct a homeomorphism $f : 2^{\mathbb{N}} \rightarrow X$. Now, it turns out that with a small change to the algorithm for constructing the I_ξ , we can arrange for $f(\overline{2^{<\mathbb{N}}})$ to be any countable dense subset of $\mathcal{C}$ that we want. Once this is proved, it will follow directly that $\mathcal{C}$ is countable dense homogeneous.

Theorem 1.28. *Let $X \neq \emptyset$ be a topological space that is T_1 , compact, has a countable base, is 0-dimensional and has no isolated points. Let E be a countable dense subset of it. Then there is a homeomorphism $f : 2^{\mathbb{N}} \rightarrow X$ with $f(\overline{2^{<\mathbb{N}}}) = E$.*

Proof. The proof is like that of 1.6, except for a couple of differences that I explain below.

Let (e_n) be an enumeration of E . Given an open subset A of X , we denote by $e(A)$ the first element of the sequence e_n that belongs to A .

When defining I_ξ for each $\xi \in 2^{<\mathbb{N}}$, in the original proof we defined $I_{\xi \wedge (0)} = I_\xi \cap B_n$ and $I_{\xi \wedge (1)} = I_\xi - B_n$. In this modified version, we look at where $e(I_\xi)$ lies. If it lies in B_n , we define $I_{\xi \wedge (0)} = I_\xi \cap B_n$ and $I_{\xi \wedge (1)} = I_\xi - B_n$, as before. If, on the other hand, $e(I_\xi) \notin B_n$, we define $I_{\xi \wedge (0)} = I_\xi - B_n$ and $I_{\xi \wedge (1)} = I_\xi \cap B_n$. Through this modification, we will ensure that each element $\bar{\xi}$ of $\overline{2^{<\mathbb{N}}}$ has image $e(I_\xi)$.

The rest of the proof that f is a homeomorphism is the same as that of 1.6: interchanging $I_{\xi \wedge (0)}$ and $I_{\xi \wedge (1)}$ has no effect. It remains to show that $f(\overline{2^{<\mathbb{N}}}) = E$.

Given $\bar{\xi} = (\xi_1, \dots, \xi_k, 0, 0, \dots) \in \overline{2^{<\mathbb{N}}}$, let us show that $f(\bar{\xi}) = e(I_\xi)$. $e(I_\xi)$ belongs to I_ξ , and by the way we have constructed $I_{\xi \wedge (0)}$, we will have $e(I_\xi) \in I_{\xi \wedge (0)}$. Therefore, since $I_{\xi \wedge (0)} \subseteq I_\xi$, $e(I_\xi)$ will be the first element of (e_n) that appears in $I_{\xi \wedge (0)}$, that is, $e(I_\xi) = e(I_{\xi \wedge (0)})$. Continuing this inductive process we see that $e(I_\xi) \in I_{\xi \wedge (0..n)}$ and $e(I_\xi) = e(I_{\xi \wedge (0..n)})$ for each n . That is, $e(I_\xi) \in I_{\bar{\xi}|_{k+n}}$ for every n , so $f(\bar{\xi}) = e(I_\xi)$. This already gives us $f(\overline{2^{<\mathbb{N}}}) \subseteq E$.

Let us now show that the image of $\overline{2^{<\mathbb{N}}}$ is all of E . Let $e_n \in E$, with $a = (a_n) = f^{-1}(e_n) \in 2^{\mathbb{N}}$. Then $e_n = \bigcap_{n \in \mathbb{N}} I_{a|_n}$. Therefore for some N , $I_{a|_N}$ contains none of the points e_i , with $i < n$. Therefore $e_n = e(I_{a|_N})$. By what we saw in the previous paragraph, this implies that $e_n = e(I_{a|_N}) = f(\overline{a|_N})$: it is the image of a point of $\overline{2^{<\mathbb{N}}}$. $\square$

Corollary 1.29. *$\mathcal{C}$ is countable dense homogeneous.*

Proof. Given countable dense subspaces D, E of $\mathcal{C}$, by 1.28 there are homeomorphisms $f_1 : 2^{\mathbb{N}} \rightarrow \mathcal{C}$ with $f_1(\overline{2^{<\mathbb{N}}}) = D$ and $f_2 : 2^{\mathbb{N}} \rightarrow \mathcal{C}$ with $f_2(\overline{2^{<\mathbb{N}}}) = E$, so $f_2 \circ f_1^{-1} = f : \mathcal{C} \rightarrow \mathcal{C}$ is a homeomorphism and $f(D) = E$. $\square$

In Appendix A one can find an even stronger homogeneity property of $\mathcal{C}$ (original, as far as I know).

Chapter 2

The space of irrationals, $\mathbb{I}$

There is no universally accepted notation for the set of irrationals, but here I will use $\mathbb{I}$. Of course, we consider $\mathbb{I}$ with its topology as a subset of $\mathbb{R}$. As with $\mathcal{C}$, in this chapter we will study some properties of $\mathbb{I}$ and characterize it topologically, and study its subsets, continuous images and quotients.

2.1 Properties of $\mathbb{I}$. Completely metrizable spaces

Just as $\mathcal{C}$ is homeomorphic to $2^{\mathbb{N}}$, $\mathbb{I}$ will also be homeomorphic to a product, in this case $\mathbb{N}^{\mathbb{N}}$. In this section we study the properties of $\mathbb{I}$ needed to characterize it. The main result of the section, 2.6, is the easy implication of Theorem 24.12 of [Wi70], and the route followed to prove it is the same.

The space $\mathbb{I}$ is metric. It also has a countable base and is 0-dimensional: a countable base of clopen sets is $\{(a, b); a, b \in \mathbb{Q}\}$. To characterize $\mathbb{I}$ we need two more properties, which we state separately.

Definition 2.1. Given a space X and one of its points x , we say that X is *locally compact at x* if x is in the interior of some compact subspace of X .¹

The irrationals will be nowhere locally compact. Intuitively, this happens because the irrationals are ‘full of holes’ (the rationals). We prove this through the following more general proposition:

Proposition 2.2. *Let X be a T_2 space, and let $A \subseteq X$ be such that A and $X - A$ are dense in X . Then A is nowhere locally compact.*

Proof. Suppose that there is a point x in the interior of a compact set $C \subseteq A$. That is, there is an open set U of X with $x \in U \cap A \subseteq C$. Since C is closed, $\overline{U \cap A} \subseteq C$. But $U \cap A$ is dense in U , so $\overline{U \cap A} = \overline{U}$. Hence $U \subseteq \overline{U} = \overline{U \cap A} \subseteq C \subseteq A$, which is impossible since $X - A$ is dense in X . $\square$

The other property has more to do with metric spaces:

Definition 2.3. A metric d on X is complete if every Cauchy sequence x_n in X converges to some point. A space X is *completely metrizable* if there is a complete metric on X that generates its topology.

If X is completely metrizable by a metric d and $f : Y \rightarrow X$ is a homeomorphism, then the metric on Y given by $d_Y(y_1, y_2) = d(f(y_1), f(y_2))$ generates the topology of Y and makes it completely metrizable. That is, being completely metrizable is a topological property. A fairly comprehensive treatment of these spaces can be found in [Wi70], Chapter 24.

For example, $\mathbb{R}$ is a complete metric space with its usual metric. Obviously, the usual metric on $\mathbb{I}$ does not make it a complete metric space. But, surprisingly, $\mathbb{I}$ is completely metrizable. We will see this as a consequence of a much more general result. First, a lemma:

¹Other definitions of local compactness are often used, but they all coincide with the one given here in T_2 spaces.

Lemma 2.4. *If X is completely metrizable and Y is an open subspace of X , Y is completely metrizable.*

Proof. Let (X, d) be complete and Y open in X . We can define the function $g : Y \rightarrow \mathbb{R}^+$; $f(y) = d(y, X - Y)$, which will be > 0 since Y is open, and continuous since, by the triangle inequality, it is Lipschitz with constant 1. Therefore the function $f(y) = \frac{1}{g(y)} = \frac{1}{d(y, X - Y)}$ will also be continuous on Y .

We define on Y the metric

$$d_Y(y_1, y_2) = d(y_1, y_2) + |f(y_1) - f(y_2)|$$

It will suffice to show that d_Y is a complete metric and generates the topology of Y .

Let us show that they generate the same topology by checking that convergence of sequences y_n in Y to a point $y \in Y$ is preserved. Since $d < d_Y$, if $d_Y(y_n, y) \rightarrow 0$, obviously $d(y_n, y) \rightarrow 0$. On the other hand, if $d(y_n, y) \rightarrow 0$, since f is continuous on Y we have that $|f(y_n) - f(y)|$ tends to 0, so $d_Y(y_n, y) \rightarrow 0$.

To show that d_Y is complete, suppose that we have a d_Y -Cauchy sequence, (y_n) . This sequence will also be d -Cauchy, so it will converge to some point $x \in X$. Moreover, x will have to be a point of Y , since otherwise we would have $f(y_n) \rightarrow \infty$. This is impossible because $f(y_n)$ is a Cauchy sequence in $\mathbb{R}$, since $|f(y_n) - f(y_m)| \leq d_Y(y_n, y_m)$. $\square$

Definition 2.5. Let Y be a subspace of X . We say that Y is a G_δ in X if Y is the intersection of a countable collection of open sets of X .

For example, in a metric space every closed subset is G_δ , and in a T_1 space every set with a countable complement is G_δ . Thus, in the particular case that interests us, $\mathbb{I}$ is a G_δ in $\mathbb{R}$.

Proposition 2.6. *Let X be a completely metrizable space and let Y be a G_δ in X . Then Y is completely metrizable.*

Proof. We denote by Y_n countably many open sets of X that satisfy $Y = \bigcap_{n \in \mathbb{N}} Y_n$. Then by 2.4, there will be metrics d_n that generate the topology of Y_n and such that (Y_n, d_n) is complete. We may assume that d_n is ≤ 1 , replacing it otherwise with the metric $\min(d_n, 1)$, which is also complete and generates the same topology.

We consider on Y the metric

$$d_Y(y_1, y_2) = \sum_{i=1}^{\infty} \frac{d_i(y_1, y_2)}{2^i}$$

It will suffice to show that d_Y is a complete metric and generates the topology of Y .

The topology on Y as a subspace of X will be the one generated by the metric d_1 (for example). Let us show that d_1 and d_Y generate the same topology by checking that convergence of sequences (y_n) in Y to a point $y \in Y$ is preserved. If $d_Y(y_n, y) \rightarrow 0$, then $d_1(y_n, y) \rightarrow 0$, since $d_1 < 2d_Y$. On the other hand, suppose that $d_1(y_n, y) \rightarrow 0$, that is, y_n converges to y in X . Then $d_i(y_n, y) \rightarrow 0$ for each $i \in \mathbb{N}$. Now, given $\varepsilon > 0$, we can take n such that $\frac{1}{2^n} < \frac{\varepsilon}{2}$, and there will be an $N > n$ such that $\forall m > N, d_i(y_m, y) < \frac{1}{n \cdot 2^n}$ for $i = 1, \dots, n$. Thus $\forall m > N$,

$$d(y_m, y) = \sum_{i=1}^n \frac{d_i(y_m, y)}{2^i} + \sum_{i=n+1}^{\infty} \frac{d_i(y_m, y)}{2^i} \leq n \frac{1}{n \cdot 2^n} + \sum_{i=n+1}^{\infty} \frac{1}{2^i} \leq \frac{2}{2^n} \leq \varepsilon.$$

Let us show that d_Y is a complete metric on Y . Since $d_n \leq 2^n d_Y$, if (y_n) is a d_Y -Cauchy sequence, then it is d_n -Cauchy for every n , so it converges to a point of Y_n , which we will call p_n . By uniqueness of the limit of the sequence (y_n) in X , all the p_n are the same point of X , p , which is the limit of y_n . Since p is in all the Y_n , it is in Y . $\square$

We have thus established that $\mathbb{I}$ is completely metrizable. The proofs of 2.4 and 2.6 are constructive, so for the curious (like me) we can explicitly give a complete metric on $\mathbb{I}$:

Proposition 2.7. *The space $\mathbb{I}$ is completely metrizable. In fact, if q_n is an enumeration of $\mathbb{Q}$, the metric on $\mathbb{I}$ given by*

$$d(i_1, i_2) = \sum_{n \in \mathbb{N}} 2^{-n} \min \left(1, |i_1 - i_2| + \left| \frac{1}{|i_1 - q_n|} - \frac{1}{|i_2 - q_n|} \right| \right)$$

is complete and generates the usual topology of $\mathbb{I}$. □

2.2 Characterization and homogeneity of $\mathbb{I}$. Baire space, $\mathbb{N}^{\mathbb{N}}$

The following theorem will prove that $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to $\mathbb{I}$ and establish the characterization of $\mathbb{I}$, as well as proving that $\mathbb{I}$ is countable dense homogeneous (1.26). The proof is taken from [Eb77]. To state it we recall some notation for $\mathbb{N}^{\mathbb{N}}$, analogous to that for $2^{\mathbb{N}}$ introduced in 1.2 and 1.27.

Notation 2.8. Baire space, $\mathbb{N}^{\mathbb{N}}$, is the space of sequences of natural numbers, $(a_n)_{n \in \mathbb{N}}$, with the product topology.

$\mathbb{N}^{<\mathbb{N}}$ is the set of finite sequences of natural numbers, $\xi = (\xi_1, \dots, \xi_k)$, including the empty sequence, $()$.

If $(a_n) \in \mathbb{N}^{\mathbb{N}}$, $a|_k$ will be the sequence of the first k terms of a : $a|_k = (a_1, \dots, a_k) \in \mathbb{N}^{<\mathbb{N}}$.

A base for the topology of $\mathbb{N}^{\mathbb{N}}$ is given by open sets of the form

$$B_\xi = \{(a_n) \in \mathbb{N}^{\mathbb{N}}; a_1 = \xi_1, \dots, a_k = \xi_k\}, \text{ for each } \xi = (\xi_1, \dots, \xi_k) \in \mathbb{N}^{<\mathbb{N}}.$$

That is, B_ξ consists of the sequences that ‘begin with ξ ’.

Given $\xi = (\xi_1, \dots, \xi_k)$, we denote by $\bar{\xi} \in \mathbb{N}^{\mathbb{N}}$ the result of completing ξ with zeros, $(\xi_1, \dots, \xi_k, 0, 0, \dots)$.

We will denote by $\overline{\mathbb{N}^{<\mathbb{N}}}$ the subset of $\mathbb{N}^{\mathbb{N}}$ consisting of its elements with finitely many nonzero entries, that is, the sequences that are 0 from some point onwards.

Theorem 2.9. *Let X be a completely metrizable, separable, 0-dimensional, nowhere locally compact space. Let E be a countable dense subset of X . Then there is a homeomorphism $h: \mathbb{N}^{\mathbb{N}} \rightarrow X$ with $h(\overline{\mathbb{N}^{<\mathbb{N}}}) = E$.*

The proof will be very similar to that of Theorem 1.6: for each ξ in $\mathbb{N}^{<\mathbb{N}}$ of length n we will choose a clopen set I_ξ of diameter $\leq \frac{1}{n}$, so that the I_ξ with ξ of length n will form a partition of X , and given (a_n) in $\mathbb{N}^{\mathbb{N}}$, $I_{a|_n}$ will be a decreasing sequence of subsets of X whose intersection is a single point. And then our homeomorphism will map (a_n) to that intersection. Some additional detail will also be needed to ensure that $h(\overline{\mathbb{N}^{<\mathbb{N}}})$ is indeed E , as in 1.29.

The following two lemmas reflect how the two new properties, being nowhere locally compact and being completely metrizable, will be used in the proof.

Lemma 2.10. *Let (X, d) be a noncompact, complete metric space that is separable and 0-dimensional. Then for each $\varepsilon > 0$ we can find a partition of X into countably infinitely many nonempty clopen sets of diameter $< \varepsilon$.*

Proof. By 1.13, X will not be totally bounded. Therefore there is some $\varepsilon_1 \leq \varepsilon$ such that there is no finite cover of X by subsets of diameter $< \varepsilon$. Since X is separable and metric it has a countable base, and by 1.5 we can find a base A_n of clopen sets of diameter $< \varepsilon_1$. Now we define $B_n = A_n - \bigcup_{i=1}^{n-1} A_i$. The B_n will be clopen, disjoint, will cover X , and infinitely many of them will be nonempty, thus proving the lemma. □

Lemma 2.11. *Let X be a complete metric space, and C_n a decreasing sequence of nonempty closed subsets whose diameters satisfy $\text{diam}(C_n) \rightarrow 0$. Then $\bigcap_{n \in \mathbb{N}} C_n$ will have exactly 1 point.*

Proof. It is clear that the intersection cannot have more than one point, since $\text{diam}(\bigcap_{n \in \mathbb{N}} C_n) \leq \text{diam}(C_n)$ for each n . On the other hand, take a sequence (x_n) with $x_n \in C_n$. Then x_n is Cauchy, and converges to some x in X . Since C_n is closed, we will have $x \in C_n$ for every n , so $x \in \bigcap_{n \in \mathbb{N}} C_n$. □

Proof of Theorem 2.9. Let $(e_n)_{n \in \mathbb{N}}$ be an enumeration of E . For each open subset A of X , we will denote by $e(A)$ the first element of the sequence (e_n) that belongs to A . We will define clopen subsets I_ξ of X , for each $\xi \in \mathbb{N}^{<\mathbb{N}}$, recursively on the length of ξ . In the base case we define $I_{()} = X$.

Now let ξ have length n and suppose that we have already constructed a nonempty clopen set I_ξ . We will construct $I_{\xi \wedge (i)}$ for $i \in \mathbb{N}$. I_ξ satisfies the conditions of 2.10, so there will be a partition of I_ξ into countably many nonempty clopen subsets of diameter $< \frac{1}{n+1}$, which will be our $I_{\xi \wedge (i)}$. We order them so that $e(I_\xi)$ lies in $I_{\xi \wedge (0)}$.

Constructed in this way, the I_ξ will have these properties:

- (1) I_ξ is a nonempty clopen set for every $\xi \in \mathbb{N}^{<\mathbb{N}}$.
- (2) For every n , $\{I_\xi; \xi \in \mathbb{N}^n\}$ is a partition of X into clopen sets of diameter $< \frac{1}{n}$.
- (3) For each $(a_n) \in \mathbb{N}^\mathbb{N}$, $I_{a|_n}$ is a decreasing sequence of closed sets whose intersection has 1 point.
- (4) $\{I_\xi; \xi \in \mathbb{N}^{<\mathbb{N}}\}$ is a base of open sets of X .

It is clear that (1) and (2) hold by induction, while in (3), to see that the intersection is 1 point, it suffices to use Lemma 2.11. (4) holds because for each point x and for each n , by (2) there is some I_ξ of diameter $< \frac{1}{n}$ that contains x .

The homeomorphism we want to construct is the following:

$$\begin{aligned} f : \quad \mathbb{N}^\mathbb{N} &\rightarrow X; \\ (a_n) &\mapsto y \quad , \text{ where } \{y\} = \bigcap_{n=1}^\infty I_{a|_n}. \end{aligned}$$

By (3), f is well-defined; it also satisfies:

- (5) Given $x \in X$, $x = f((a_n))$, where $a = (a_n)$ is the sequence such that $I_{a|_k}$ contains x for each k .

The (a_n) in (5) can be constructed recursively, with a_n being the unique i such that x lies in $I_{a|_{n-1} \wedge (i)}$.

(5) makes it clear that f is surjective. It is injective since, given distinct $a = (a_n)$ and $b = (b_n)$ in $\mathbb{N}^\mathbb{N}$, there will be some k such that $a|_k \neq b|_k$, so by (2), $I_{a|_k}$ and $I_{b|_k}$ will be disjoint. Therefore $f(a) \neq f(b)$.

To show that f is a homeomorphism it would suffice to show that $f(B_\xi) = I_\xi$ for every $\xi \in \mathbb{N}^{<\mathbb{N}}$, since $\{B_\xi\}_{\xi \in \mathbb{N}^{<\mathbb{N}}}$ is a base for the topology of $\mathbb{N}^\mathbb{N}$ (2.8) and by (4) $\{I_\xi\}_{\xi \in \mathbb{N}^{<\mathbb{N}}}$ is a base for the topology of X . But $f(B_\xi) = I_\xi$ for every ξ follows directly from the definition of f and (5).

Finally, let us show that $f(\overline{\mathbb{N}^{<\mathbb{N}}}) = E$.

Given $\bar{\xi} = (\xi_1, \dots, \xi_k, 0, 0, \dots) \in \overline{\mathbb{N}^{<\mathbb{N}}}$, let us show that $f(\bar{\xi}) = e(I_\xi)$. It is clear that $e(I_\xi)$ belongs to I_ξ , and by the way we have constructed the I_ξ , $e(I_\xi) \in I_{\xi \wedge (0)}$. Therefore, since $I_{\xi \wedge (0)} \subseteq I_\xi$, $e(I_\xi)$ will be the first element of (e_n) that appears in $I_{\xi \wedge (0)}$, that is, $e(I_\xi) = e(I_{\xi \wedge (0)})$. Continuing this inductive process we see that $e(I_\xi) \in I_{\xi \wedge (0, \dots, 0)}$ and $e(I_\xi) = e(I_{\xi \wedge (0, \dots, 0)})$ for each n . That is, $e(I_\xi) \in I_{\xi|_{k+n}}$ for every n , so $f(\bar{\xi}) = e(I_\xi)$. This already gives us $f(\overline{\mathbb{N}^{<\mathbb{N}}}) \subseteq E$.

Now given $e_n \in E$, there is some $\varepsilon > 0$ such that $d(e_n, e_m) > \varepsilon$ for each $m < n$. Therefore, if we take ξ of length n such that $e_n \in I_\xi$, $e_0, \dots, e_{n-1}$ will not be in I_ξ . Therefore $e_n = e(I_\xi)$. By what we saw in the previous paragraph, this implies that $e_n = f(\bar{\xi})$, hence $E \subseteq f(\overline{\mathbb{N}^{<\mathbb{N}}})$. $\square$

Corollary 2.12. *X is homeomorphic to $\mathbb{I}$ (and to $\mathbb{N}^\mathbb{N}$) if and only if it is a completely metrizable, separable, 0-dimensional, nowhere locally compact space.* $\square$

Corollary 2.13. *$\mathbb{I}$ is countable dense homogeneous.*

Proof. Given countable dense subspaces D, E of $\mathbb{I}$, by 2.9 there are homeomorphisms $f_1 : \mathbb{N}^\mathbb{N} \rightarrow \mathbb{I}$ with $f_1(\overline{\mathbb{N}^{<\mathbb{N}}}) = D$ and $f_2 : \mathbb{N}^\mathbb{N} \rightarrow \mathbb{I}$ with $f_2(\overline{\mathbb{N}^{<\mathbb{N}}}) = E$, so $f_2 \circ f_1^{-1} = f : \mathbb{I} \rightarrow \mathbb{I}$ is a homeomorphism and $f(D) = E$. $\square$

I find Corollary 2.13 very surprising; it tells us, for example, that there is a homeomorphism $f : \mathbb{I} \rightarrow \mathbb{I}$ with $f(\pi + \mathbb{Q}) = \pi + \mathbb{A}$, where $\mathbb{A}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{R}$, that is, the real numbers x such that there is some polynomial $p \in \mathbb{Q}[x]$ with $p(x) = 0$. In Appendix A we see an even stronger homogeneity property of $\mathbb{I}$.

2.3 Continued fractions

In this section I informally discuss an intriguing topic that must be mentioned when discussing $\mathbb{I}$ and $\mathbb{N}^{\mathbb{N}}$: continued fractions. The details of this construction, which can be found, for example, in Chapter 10 of [Tr97], are very tedious, and including them here would be a digression.

Given a finite sequence of k natural numbers, $(a_1, a_2, \dots, a_k)$, which we will call a *continued fraction of order k* , we can associate a number with it:

$$\theta(a_1, a_2, \dots, a_k) = a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_k}}}$$

In fact, given a sequence of natural numbers, $a = (a_n)_{n \in \mathbb{N}}$, the sequence $\theta(a|_k) = \theta(a_1, a_2, \dots, a_k)$ will always converge as k tends to infinity, so that we obtain a number associated with a :

$$\theta(a) = \lim_{k \rightarrow \infty} \theta(a|_k) = a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}$$

And it turns out that this limit will always be an irrational number (unlike continued fractions of finite order, which will obviously have rational numbers associated with them).

Looking at this process in reverse, if we start with an irrational number $x > 1$, we can obtain a continued fraction $a = (a_n)$ with $\theta(a) = x$. Indeed, from the equation

$$x = a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}$$

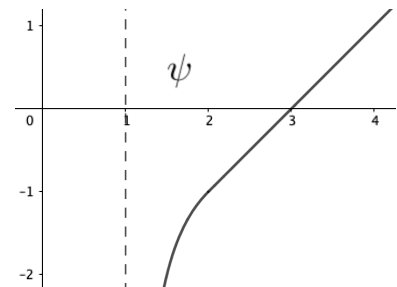
We obtain $a_1 = \lfloor x \rfloor$, since the remainder of the continued fraction is < 1 . Rearranging, we find that

$$\frac{1}{x - a_1} = a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}$$

so that $a_2 = \left\lfloor \frac{1}{x - a_1} \right\rfloor$. In this way the continued fraction can be determined recursively from x .

In number theory, continued fractions are important because the sequence of continued fractions of a number x is ‘the best sequence of approximations’ to x by rational numbers: if (a_n) is the continued fraction associated with x , and we write $\theta(a|_k) = \frac{a_k}{b_k}$, then $\left(\frac{a_k}{b_k}\right)$ converges to x , but moreover, for each k , $\frac{a_k}{b_k}$ is the rational number closest to x among those with denominator $\leq b_k$. Furthermore, $|b_k x - a_k|$ is the smallest value of $|bx - a|$ for any fraction $\frac{a}{b}$ with denominator $\leq b_k$.

Returning to the topic at hand, continued fractions will give us a bijection θ from the sequences of natural numbers, $\mathbb{N}^{\mathbb{N}}$, to $\mathbb{I} \cap (1, \infty)$. What is surprising is that, as shown in Chapter 10 of [Tr97], θ is in fact a homeomorphism. Moreover, it is easy to find an explicit homeomorphism between $\mathbb{I}$ and $\mathbb{I} \cap (1, \infty)$:



$$\psi : \mathbb{I} \cap (1, \infty) \rightarrow \mathbb{I}; x \rightarrow \begin{cases} \frac{1}{1-x} & \text{if } x \in (1, 2) \\ x-3 & \text{if } x > 2 \end{cases}$$

Thus $\psi \circ \theta$ is an ‘explicit’ homeomorphism from $\mathbb{N}^{\mathbb{N}}$ to $\mathbb{I}$.

2.4 Subspaces, images and quotients of $\mathbb{I}$. Polish and analytic spaces

It will be very easy to determine which spaces are homeomorphic to a subspace of $\mathbb{I}$. To do so, we first note that $2^{\mathbb{N}}$ is a subspace of $\mathbb{N}^{\mathbb{N}}$. Indeed, it is straightforward to check that the inclusion $2^{\mathbb{N}} \hookrightarrow \mathbb{N}^{\mathbb{N}}$ is a homeomorphism onto its image.

Proposition 2.14. *A space X is homeomorphic to a subset of $\mathbb{I}$ if and only if it is T_0 , has a countable base and is 0-dimensional.*

Proof. By what we have just seen, $2^{\mathbb{N}}$ is homeomorphic to a subspace of $\mathbb{N}^{\mathbb{N}}$. Moreover, by Theorem 1.16, $\mathbb{N}^{\mathbb{N}}$ is homeomorphic to a subspace of $2^{\mathbb{N}}$. Therefore, if X is homeomorphic to a subspace of $2^{\mathbb{N}}$, it will be homeomorphic to one of $\mathbb{N}^{\mathbb{N}}$, and vice versa. Thus by 1.16 we are done. $\square$

We move on to study the images of $\mathbb{I}$.

Definition 2.15. A *Polish space* is a completely metrizable and separable space.

As we will now see, all Polish spaces are images of $\mathbb{I}$. However, the converse is not true, even if we impose separation or countability conditions: $\mathbb{Q}$ is a continuous image of $\mathbb{I}$ and is not a Polish space (since it can be shown that every nonempty completely metrizable space without isolated points has at least $2^{\aleph_0}$ points). To find a continuous surjective function $f : \mathbb{I} \rightarrow \mathbb{Q}$, it suffices, given an enumeration q_n of $\mathbb{Q}$ and a partition C_n of $\mathbb{I}$ into infinitely many disjoint clopen sets, to define $f(C_n) = \{q_n\}$.

Theorem 2.16. *If X is a Polish space, there is a continuous surjective function $f : \mathbb{I} \rightarrow X$.*

Proof. The proof will be analogous to that of the Hausdorff-Alexandroff theorem 1.12. Let d be a complete metric on X .

We will construct a family of nonempty subsets of X , $\{F_\xi\}_{\xi \in \mathbb{N}^{<\mathbb{N}}}$, so that:

- (1) $F_\emptyset = X$
- (2) $F_\xi = \bigcup_{n \in \mathbb{N}} F_{\xi \wedge (n)}$
- (3) If ξ has length n , $\text{diam}(F_\xi) < \frac{1}{n}$.

To construct it recursively it suffices to note that, since X has a countable base, F_ξ has a countable base for each ξ of length n , so it is Lindelöf, hence it has a cover by countably many balls of diameter $< \frac{1}{n+1}$, which will be our $F_{\xi \wedge (i)}$, with $i \in \mathbb{N}$.

Using these properties, it follows directly by induction on n that the F_ξ , with ξ of length n , form a cover of X by subsets of diameter $< \frac{1}{n}$.

Now, given a sequence $s \in \mathbb{N}^{\mathbb{N}}$, we can associate with it the unique point x_s of $\bigcap_{n \in \mathbb{N}} \overline{F_{s|_n}}$ (this set has a unique point by 2.11). We proceed to define the function f we are looking for:

$$\begin{aligned} f : \quad \mathbb{N}^{\mathbb{N}} &\rightarrow X; \\ s = (s_n)_{n \in \mathbb{N}} &\mapsto x_s \end{aligned}$$

This function is surjective, since given x , we can recursively construct a sequence $s = (s_n)$ such that $x \in F_{s|_n}$ for every n . It is also continuous, since given s and a ball $B(x_s, \frac{1}{n})$, we have that all sequences (a_n) with $a|_n = s|_n$ (a neighborhood of s in $\mathbb{N}^{\mathbb{N}}$) will satisfy $f(a) \in B(x_s, \frac{1}{n})$. $\square$

Although we will not be able to characterize the images of $\mathbb{I}$ by simple properties as we have done in other cases, the metric images of $\mathbb{I}$ are extensively studied sets in descriptive set theory: they are known as analytic sets. A very detailed study of them, and of Polish spaces in general, can be found in [Ke95].

Definition 2.17. Given a Polish space X , we say that a subset $A \subseteq X$ is analytic if there is a Polish space Y and a continuous function $f : Y \rightarrow X$ with $f(Y) = A$.

Theorem 2.18. *Given a nonempty metric space X , X is a continuous image of $\mathbb{I}$ if and only if it is homeomorphic to an analytic subset of a Polish space.*

Proof. If X is homeomorphic to an analytic subset of a Polish space, then it is the continuous image of a Polish space. Since Polish spaces are in turn continuous images of $\mathbb{I}$, this proves this implication.

Conversely, if X is a continuous image of $\mathbb{I}$, we can view X as a subspace of its completion $\overline{X}$, which is separable (since X is), and is therefore a Polish space. Since X is a continuous image of $\mathbb{I}$, it is an analytic subset of $\overline{X}$. $\square$

Turning to the quotients of $\mathbb{I}$, we will prove that, in fact, all continuous images of the irrationals that are metric spaces are quotients of the irrationals. That is, the metric quotients of the irrationals will also be, up to homeomorphism, the analytic subspaces of Polish spaces. Theorem 2.20, and its proof, are taken from [MS69]. To prove it we will need this lemma:

Lemma 2.19. *If $X \neq \emptyset$ is a metrizable and separable space, X has a metric d and a base of nonempty open sets $(V_n)_{n \in \mathbb{N}}$ such that $\text{diam}(V_n) \rightarrow 0$ and each $x \in X$ belongs to infinitely many distinct V_i .*

Proof. We begin by showing that X is homeomorphic to a subspace of the Hilbert cube, $\mathcal{H}$, defined as the topological product $[0, 1]^{\mathbb{N}}$ (this can be seen, for example, in 23.1 of [Wi70]).

To do so, we consider (e_n) a dense sequence in X , and a metric $d_1 \leq 1$ on X , and consider the function

$$\begin{aligned} f : X &\rightarrow \mathcal{H}; \\ x &\mapsto (d_1(x, e_n))_{n \in \mathbb{N}}. \end{aligned}$$

f is continuous since, if x_i tends to x , then $d_1(x_i, e_n)$ tends to $d_1(x, e_n)$ for every n , so $f(x_i)$ tends to $f(x)$. It is also open onto its image, since given a ball $B := B(x, \varepsilon)$, and some e_n at distance $< \frac{\varepsilon}{2}$ from x , then $f(B)$ contains the elements of $f(X)$ whose n th coordinate is $< \frac{\varepsilon}{2}$, which form an open subset of the image containing $f(x)$. That f is injective follows easily from the triangle inequality and the fact that (e_n) is dense in X , so indeed f is a homeomorphism onto its image.

Thus we may assume that X is a subset of $\mathcal{H}$, which is metrizable and compact, being a countable product of metrizable compact spaces, and we give X the metric d inherited from that of $\mathcal{H}$. Now, since $\mathcal{H}$ is totally bounded, let $\mathcal{U}_n$ be a finite cover of $\mathcal{H}$ by open sets of diameter $< \frac{1}{n}$. Let (U_n) be an enumeration of $\cup_{i=1}^{\infty} \mathcal{U}_n$. Then $\text{diam}(U_n) \rightarrow 0$ and each $x \in \mathcal{H}$ belongs to infinitely many U_i . We can obtain the base V_n in the statement by taking $V_n = U_n \cap X$, and omitting the empty open sets that remain in the sequence. $\square$

Theorem 2.20. *If a metrizable space X is a continuous image of $\mathbb{I}$, then X is a quotient of $\mathbb{I}$.*

Proof. Since X is separable, we give it a metric d and a base V_n as indicated in the preceding lemma.

Our strategy will consist of constructing a subspace Y of the plane homeomorphic to $\mathbb{I}$ and a quotient map $f : Y \rightarrow X$.

Let $g : \mathbb{I} \rightarrow X$ be a continuous function from $\mathbb{I}$ onto X . We define $Y_0 = \mathbb{I} \times \{0\}$.

For each $n \in \mathbb{N}$ and $j \in \mathbb{Z}$, we define $A_{nj} = g^{-1}(V_n) \cap [\frac{j}{n}, \frac{j+1}{n}]$, and $Y_{nj} = A_{nj} \times \{\frac{1}{n}\}$.

Now, writing $J_n = \{j; A_{nj} \neq \emptyset\}$, we can define Y :

$$Y = Y_0 \cup \bigcup \{Y_{nj}; n \in \mathbb{N}; j \in J_n\}$$

By their definition, Y_0 and the Y_{nj} are all pairwise disjoint, and the Y_{nj} are clopen in Y . However, the Y_{nj} are homeomorphic to the A_{nj} , nonempty open subsets of $\mathbb{I}$, and therefore homeomorphic to $\mathbb{I}$. For the same

reason, $g^{-1}(V_n)$ is homeomorphic to $\mathbb{I}$. Since Y_{nj} and $g^{-1}(V_n)$ are homeomorphic, there will be a continuous surjective function $f_{nj} : Y_{nj} \rightarrow V_n$.

We define $f : Y \rightarrow X$ so that $f|_{Y_{nj}} = f_{nj}$ for $n \in \mathbb{N}, j \in J_n$, and on Y_0 we define $f(s, 0) = g(s)$. To prove the result we have to check three things:

- Y is homeomorphic to $\mathbb{I}$.

To check this, of course, we will use Theorem 2.12. Y is separable, metric and has no isolated points. Y will be nowhere locally compact: at points of the Y_{nj} this is immediate because the Y_{nj} are clopen sets homeomorphic to $\mathbb{I}$, and a point y of Y_0 cannot have a compact neighborhood U , since then $U \cap Y_0$, which is homeomorphic to an open subset of $\mathbb{I}$, would be compact (since Y_0 is closed in Y). It only remains to show that Y admits a complete metric. But $\mathbb{I}$ is completely metrizable, so $\mathbb{I} \times \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$ is as well. Moreover, Y is a G_δ in that set, since it is obtained by removing a closed set from each $\mathbb{I} \times \{\frac{1}{n}\}$. Therefore by 2.6 Y is completely metrizable.

- f is continuous.

We check this at each point. It is obvious at the points of the clopen sets Y_{nj} . Let us check it on Y_0 .

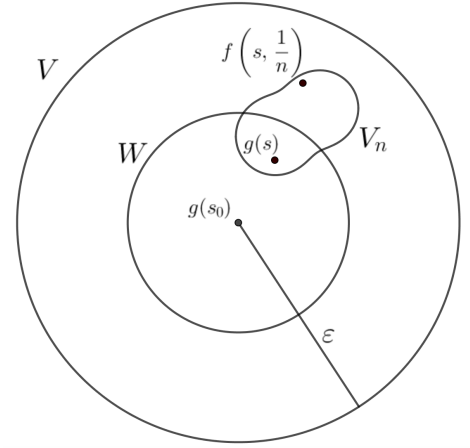
Let $(s_0, 0) \in Y_0$, and let $V = B(g(s_0), \varepsilon) \subseteq X$ be a neighborhood of its image. Let $W = B(g(s_0), \frac{\varepsilon}{2})$ and let n_0 be such that $\text{diam}(V_n) < \frac{\varepsilon}{2}$ for every $n \geq n_0$. Then, writing $U = Y \cap \left(g^{-1}(W) \times \left[0, \frac{1}{n_0}\right]\right)$, we will check that $f(U) \subseteq V$.

If $(s, 0) \in U$, then $f((s, 0)) \in W \subseteq V$. On the other hand, if $(s, \frac{1}{n}) \in U$, then $f(s, 0)$ lies in W and in V_n , and $f(s, \frac{1}{n})$ lies in V_n . Therefore $d(f(s_0, 0), f((s, \frac{1}{n}))) \leq d(f(s_0, 0), f(s, 0)) + d(f(s, 0), f((s, \frac{1}{n}))) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \leq \varepsilon$. That is, $f((s, \frac{1}{n})) \in V$.

- f is a quotient map.

Let $x \in X$ and let U be a neighborhood in Y of $f^{-1}(x)$. Then, choosing $y \in \mathbb{I}$ with $g(y) = x$, $(y, 0) \in U$, so there is some $\varepsilon > 0$ with $(y - \varepsilon, y + \varepsilon) \times [0, \varepsilon] \subseteq U$. Now let $N > \frac{1}{\varepsilon}$ be such that $x \in V_N$. Then there is a $Y_{N,j}$ contained in $(y - \varepsilon, y + \varepsilon) \times [0, \varepsilon]$, so $V_N = f(Y_{N,j}) \subseteq f(U)$. Since V_N is a neighborhood of x , we are done.

□



Chapter 3

The space of rationals, $\mathbb{Q}$

In this chapter we discuss the surprising characterization of $\mathbb{Q}$: Sierpinski's theorem. We then obtain some results on subspaces of $\mathbb{Q}$, taking advantage of the fact that it is 'only' countable, and a topological invariant that is particularly useful for distinguishing subspaces of $\mathbb{Q}$. Finally, we characterize the images of $\mathbb{Q}$ and obtain some results on spaces that are quotients of $\mathbb{Q}$.

3.1 Sierpinski's theorem

We now have the tools needed to prove Sierpinski's theorem, which characterizes $\mathbb{Q}$. The following proof of the result is similar to that in [Eb77], except that here we take a shortcut by using the countable dense homogeneity of $2^{\mathbb{N}}$ (1.29) instead of that of $\mathbb{N}^{\mathbb{N}}$ (2.13).

Theorem 3.1 (Sierpinski). *Let $X \neq \emptyset$ be a countable metrizable space without isolated points. Then X is homeomorphic to $\mathbb{Q}$.*

Proof. First, X is T_0 . Moreover, the set $D = \{d(x, y); x, y \in X\}$ is countable, so there is a sequence of positive numbers $d_n \rightarrow 0$ whose elements do not belong to D . Thus $\{B(x, d_n); x \in X; n \in \mathbb{N}\}$ is a countable base of clopen sets of X . Therefore by 1.16 we may assume that X is a subspace of $2^{\mathbb{N}}$. Its closure $\overline{X} \subseteq 2^{\mathbb{N}}$ will be compact, separable, 0-dimensional (since $2^{\mathbb{N}}$ is) and without isolated points (since X has none). Thus by 1.6, $\overline{X}$ is homeomorphic to $2^{\mathbb{N}}$.

Thus X will be homeomorphic to a countable dense subspace of $2^{\mathbb{N}}$. Therefore, since by 1.29 all countable dense subspaces of $2^{\mathbb{N}}$ are homeomorphic, we have that all countable metrizable spaces without isolated points are homeomorphic, thus concluding the proof, since $\mathbb{Q}$ has these properties. $\square$

Theorem 3.2. *If X is a separable metrizable space without isolated points and A is a countable dense subset of X , then A is homeomorphic to $\mathbb{Q}$.*

Proof. To prove this it suffices to show that A will have no isolated points. Given a point a , since it is not isolated in X there will be a sequence x_n of points distinct from a with $x_n \rightarrow a$. In turn, since A is dense in X , there will be a sequence a_n with $d(a_n, x_n) < d(a, x_n)$. Therefore the a_n are distinct from a and $a_n \rightarrow a$, that is, A has no isolated points. $\square$

List of some spaces that are surprisingly homeomorphic to $\mathbb{Q}$

- The set of endpoints of the intervals in the C_n used in the construction of the Cantor set (which is dense in $\mathcal{C}$). Or, in general, any countable dense subspace of $\mathcal{C}$.
- $\mathbb{Q}^2$, or in general $\mathbb{Q}^n$ with $n > 1$.
- $\mathbb{A}$, the set of algebraic numbers, or any other countable field extension of $\mathbb{Q}$ in $\mathbb{C}$.
- $\mathbb{Q} \cap [0, 1]$. The endpoints are indistinguishable from the rest!

- The space $\mathbb{Q}[x]$ of polynomials with rational coefficients, with metric $d(p, q) = \max_{x \in [0, 1]} |p(x) - q(x)|$.
- $\mathbb{Q}$ with the topology whose base consists of intervals of the form $[a, b]$, with $a, b \in \mathbb{Q}$, has a countable base and is regular, so by Urysohn's Theorem 0.1 it is metrizable. Hence, since it has no isolated points, it is homeomorphic to $\mathbb{Q}$.
- $\mathbb{Z}$ with the topology whose base of clopen sets consists of the arithmetic progressions, $N_{(a, b)} = \{a + nb; n \in \mathbb{Z}\}$ with $a, b \in \mathbb{Z}, b \neq 0$, is also regular, has a countable base and has no isolated points, so it is homeomorphic to $\mathbb{Q}$.

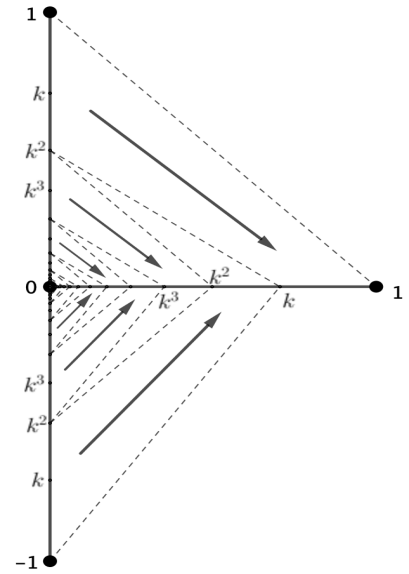
All these spaces have something in common: if, before learning about Sierpinski's theorem, I had been asked whether they are homeomorphic to $\mathbb{Q}$, I would have answered: 'Certainly not, let us see how to prove it'. In fact, I discovered this theorem while trying to prove with a fellow student that $\mathbb{Q} \cap (0, 1)$ is not homeomorphic to $\mathbb{Q} \cap [0, 1)$. The theorem tells us that they are homeomorphic, but at the time I could not think of any explicit homeomorphism. However, after seeing the idea used in the proofs of characterizations of 0-dimensional spaces, which consists of dividing the set into clopen sets and finding homeomorphisms between those clopen sets (and repeating this process), it is not difficult to find an explicit homeomorphism between the two spaces. Below I present an original one. In the following paragraphs I write $(a, b)_{\mathbb{Q}}$ for $\mathbb{Q} \cap (a, b)$.

First we need to find an explicit homeomorphism $h : (a, b)_{\mathbb{Q}} \rightarrow (c, d)_{\mathbb{Q}}$, where $b > a$ and $d > c$ are real numbers. To do this it suffices to find strictly decreasing sequences of rational numbers $a_n \rightarrow a$, $c_n \rightarrow c$, and strictly increasing sequences of rational numbers $b_n \rightarrow b$ and $d_n \rightarrow d$. We may assume $a_1 < b_1$ and $c_1 < d_1$, omitting some terms of the sequence otherwise. Now we simply define $h(a_n) = c_n$ and $h(b_n) = d_n$ and complete h by linear interpolation.

Therefore it will suffice to find an explicit homeomorphism $f : (-1, 1)_{\mathbb{Q}} \rightarrow [0, 1)_{\mathbb{Q}}$. Let $k \in (0, 1)$ be such that k^n is irrational for every n . We will define f piecewise. On $(k^2, 1)_{\mathbb{Q}}$, we define f as a homeomorphism between $(k^2, 1)_{\mathbb{Q}}$ and $(k, 1)_{\mathbb{Q}}$. On $(-1, -k^2)_{\mathbb{Q}}$, we define it as a homeomorphism between $(-1, -k^2)_{\mathbb{Q}}$ and $(k^2, k)_{\mathbb{Q}}$. We continue this process by defining f on $(k^{2n+2}, k^{2n})_{\mathbb{Q}}$ as a homeomorphism onto $(k^{2n+1}, k^{2n})_{\mathbb{Q}}$ and on $(-k^{2n+2}, -k^{2n+2})_{\mathbb{Q}}$ as a homeomorphism onto $(k^{2n+2}, k^{2n+1})_{\mathbb{Q}}$.

In the figure, where $\mathbb{Q} \cap (-1, 1)$ appears vertically and $[0, 1)_{\mathbb{Q}}$ horizontally, arrows show the images of the intervals under f . Finally, of course, we define $f(0) = 0$.

It is clear that f is a bijection, and it is continuous at every point except 0 since it is continuous on the intervals of the form $(k^{2n+2}, k^{2n})_{\mathbb{Q}}$ and $(-k^{2n+2}, -k^{2n})_{\mathbb{Q}}$. Likewise, f^{-1} is continuous at every point except 0 since it is continuous on each interval $(k^{n+1}, k^n)_{\mathbb{Q}}$. Moreover, f and its inverse are continuous at 0, since for every $x \in (-1, 1)_{\mathbb{Q}} \setminus \{0\}$, $k^2 < \frac{d(0, f(x))}{d(0, x)} < \frac{1}{k^2}$. Therefore f is a homeomorphism.



3.2 Subspaces of $\mathbb{Q}$: countable metric spaces

Using Sierpinski's theorem, it is easy to prove a characterization of the subspaces of $\mathbb{Q}$:

Theorem 3.3. *A topological space X is homeomorphic to a subspace of $\mathbb{Q}$ if and only if it is metrizable and countable.*

Proof. One implication is obvious. For the other, let (X, d) be a countable metric space. We consider the topological product $X \times \mathbb{Q}$. This space is metrizable, countable and has no isolated points, since $\mathbb{Q}$ has no isolated points. Therefore by 3.1 $X \times \mathbb{Q}$ is homeomorphic to $\mathbb{Q}$. Thus, since X is homeomorphic to $X \times \{0\}$, it will be homeomorphic to a subspace of $\mathbb{Q}$. $\square$

¹One possible explicit construction would be: if a is rational, we choose $a_n = a - \frac{1}{n}$, and if a is irrational, we choose a_n as the number with the first n nonzero decimal digits of a .

Corollary 3.4. *Every metrizable and countable space X is homeomorphic to a closed subspace of $\mathbb{Q}$. $\square$*

In the rest of the section we will exhibit uncountably many of these classes, constructed explicitly from the countable ordinals. Moreover, we will see that every countable metric space is scattered (3.9) or the union of a scattered subspace and another homeomorphic to $\mathbb{Q}$. Intuitively, this means that if we start with a countable metric space X and begin removing isolated points one by one, eventually X will be empty or we will be left with a subspace of X homeomorphic to $\mathbb{Q}$ (from which, obviously, we will no longer be able to remove isolated points). The development of this section is partly original, based on my solution to Problem 13E of [Wi70]. Moreover, some of the notation is taken from [Gi05], and the idea of Theorem 3.10 is essentially the same as that of the Cantor Bendixon theorem 6.4 in [Ke95].

M_0	$=$	$\{0\}$	M_0	$\bullet$
M_1	$=$	$\{0\} \cup \{\frac{1}{2^n}\}_{n=1}^\infty$	M_1	$\bullet \dots \bullet \quad \bullet \quad \bullet \quad \bullet$
M_2	$=$	$\{0\} \cup \bigcup_{n=1}^\infty \{\frac{1}{2^n} + \frac{M_1}{2^{n+1}}\}$	M_2	$\bullet \dots \bullet \quad \bullet \dots \bullet \quad \bullet \dots \bullet \quad \bullet \dots \bullet$
M_3	$=$	$\{0\} \cup \bigcup_{n=1}^\infty \{\frac{1}{2^n} + \frac{M_2}{2^{n+1}}\}$	M_3	$\bullet \dots \bullet \quad \bullet \dots \bullet \quad \bullet \dots \bullet \quad \bullet \dots \bullet \quad \bullet \dots \bullet$
$\vdots$				
M_k	$=$	$\{0\} \cup \bigcup_{n=1}^\infty \{\frac{1}{2^n} + \frac{M_{k-1}}{2^{n+1}}\}$		

Definition 3.5. For each countable ordinal α , we define M_α recursively as follows:

$$\begin{cases} M_0 &= \{0\} \\ M_{\alpha+1} &= \{0\} \cup \bigcup_{n=1}^{\infty} \left\{ \frac{1}{2^n} + \frac{M_\alpha}{2^{n+1}} \right\} \\ M_\alpha &= \{0\} \cup \bigcup_{n=1}^{\infty} \left\{ \frac{1}{2^n} + \frac{M_{\beta_n}}{2^{n+1}} \right\} \text{ if } \alpha \text{ is a limit ordinal, where } \beta_n \text{ is an enumeration of } \alpha. \end{cases}$$

Definition 3.6. Given a space X , we define its α -th Cantor-Bendixon derivative $X^{(\alpha)}$ for each ordinal α by induction as follows:

$$\begin{cases} X^{(0)} &= X \\ X^{(\alpha+1)} &= (X^{(\alpha)})^{(1)} \\ X^{(\alpha)} &= \bigcap_{\beta < \alpha} X^{(\beta)} \text{ If } \alpha \text{ is a limit ordinal.} \end{cases}$$

It is clear that for each X , X^α is decreasing, that is, X^β is contained in X^α if $\beta > \alpha$. Moreover, if X and Y are homeomorphic, then it follows directly by induction that X^α is homeomorphic to Y^α for every ordinal α . Another property that follows directly by induction is that if Y is an open subspace of X , then $Y^\alpha = Y \cap X^\alpha$.

If X is a countable space, then for some countable ordinal we will have $X^\alpha = X^{\alpha+1}$, since otherwise we would have, for each countable ordinal α , a point $x_\alpha \in X^{\alpha+1} - X^\alpha$, so X would be uncountable. Similarly, it can be seen that for any X , there is an ordinal α of cardinality $\leq |X|$ such that $X^\alpha = X^{\alpha+1}$. We call $N(X)$, the *Cantor-Bendixon rank of X* , the smallest ordinal that satisfies this, that is, the smallest ordinal such that $X^{N(X)}$ has no isolated points. Clearly, we will have $X^{(\alpha)} = X^{N(X)}$ if $\alpha > N(X)$. If X, Y are homeomorphic via $f : X \rightarrow Y$, since $f(X^\alpha) = Y^\alpha$ for each α , we will have that $N(X)$ is homeomorphic to $N(Y)$. We will also say that a point x is α -isolated if $x \in X^{(\alpha)} - X^{(\alpha+1)}$. For example, 0-isolated points are simply isolated points, and 0 is a 1-isolated point of M_1 . The following proposition will show that the M_α are not homeomorphic to one another:

Proposition 3.7. $M_\alpha^{(\alpha)} = \{0\}$. Therefore $N(M_\alpha) = \alpha + 1$ for every α .

Proof. Of course, we have to use induction on α . The case 0 holds. Let us consider the limit case first.

If α is a limit ordinal, M_α consists of copies of M_{β_n} for $\beta_n < \alpha$, which we will call $A_{\beta_n} = \frac{1}{2^n} + \frac{M_{\beta_n}}{2^{n+1}}$. A_{β_n} is contained in $[\frac{1}{2^n}, \frac{3}{2^{n+1}}]$, so it is clopen in M_α , being at a positive distance from $M_\alpha \setminus A_{\beta_n}$. Therefore, by the induction hypothesis, in $M_{\alpha}^{(\beta_n)}$ there will be only one point of A_{β_n} , which will be $\frac{1}{2^n}$.

This implies that $M_\alpha^{(\alpha)}$ cannot have any points other than 0, since for each $\beta < \alpha$, $M_\alpha^{(\beta)}$ will contain no point of A_β . Moreover, for each $\beta < \alpha$, there are infinitely many ordinals β_n with $\beta < \beta_n < \alpha$, and these ordinals will satisfy that $M_\alpha^{(\beta)}$ has points in A_{β_n} . Since $M_\alpha^{(\beta)}$ has points in infinitely many intervals of the form $[\frac{1}{2^n}, \frac{3}{2^{n+1}}]$ and is closed, we will have $0 \in M_\alpha^{(\beta)}$. Since this holds for every $\beta < \alpha$, $0 \in M_\alpha^{(\alpha)}$.

We move on to the successor case. Suppose that $M_\alpha^{(\alpha)} = \{0\}$. $M_{\alpha+1}$ consists of countably many homeomorphic copies of M_α that are clopen (like the A_{β_n} above), together with 0. Therefore $M_{\alpha+1}^{(\alpha)}$ will contain only the zeros of the copies of M_α , that is, the points $\frac{1}{2^n}$, and 0, since $M_{\alpha+1}^{(\alpha)}$ is closed. Hence $M_{\alpha+1}^{(\alpha+1)} = (M_{\alpha+1}^{(\alpha)})^{(1)} = \{0\}$.

Knowing that $M_\alpha^{(\alpha)} = \{0\}$, it is clear that $M_\alpha^{(\alpha+1)} = \emptyset$, so $N(M_\alpha) = \alpha + 1$. $\square$

Thus the M_α all have different Cantor-Bendixon ranks, so they cannot be homeomorphic to one another. Therefore we would have found $\aleph_1$ topologically distinct countable metric spaces. In fact, $2^{\aleph_0}$ can be found (see Appendix B). Moreover, from what we saw in the preceding proof it is clear that $N(M_\alpha - \{0\}) = \alpha$, so for each countable ordinal α there will be a space X with $N(X) = \alpha$.

Proposition 3.8. If X is countable and metric, then $X^{N(X)}$ is $\emptyset$ or homeomorphic to $\mathbb{Q}$.

Proof. $X^{N(X)}$ is metric and has no isolated points. Therefore, if it is nonempty, it will have infinitely many points. Being countable, metric and without isolated points, it will be homeomorphic to $\mathbb{Q}$. $\square$

It is easy to characterize the spaces X such that $X^{N(X)} = \emptyset$:

Proposition 3.9. $X^{N(X)} = \emptyset$ if and only if X is scattered, that is, every nonempty subset of X has isolated points.

Proof. If X is scattered, since $X^{(N(X))}$ is a subset of X without isolated points, it will have to be $\emptyset$. Conversely, suppose $X^{(N(X))} = \emptyset$, and let A be a nonempty subset of X . If A had no isolated points, then we see by induction that $A \subseteq X^{(\alpha)}$ for every α , so $A \subseteq X^{(N(X))} = \emptyset$, a contradiction. $\square$

We can now state the fact we wanted about countable metric spaces:

Proposition 3.10. *If X is countable and metric, then it is scattered or is the union of a scattered subspace and a subspace homeomorphic to $\mathbb{Q}$.*

Proof. If $X^{(N(X))} = \emptyset$, X is scattered by 3.9.

Otherwise, by 3.8, $X^{(N(X))}$ will be homeomorphic to $\mathbb{Q}$. Since $X^{(N(X))}$ is closed in X , we will have that $Y := X - X^{(N(X))}$ is open, so $Y^{(N(X))} = Y \cap X^{(N(X))} = \emptyset$. Therefore by 3.9, Y will be scattered. $\square$

A much more detailed study of countable metric spaces can be found in [Gi05].

3.3 Continuous images and quotients of $\mathbb{Q}$. Sequential spaces

$\mathbb{Q}$ is a countable space, and is itself a countable union of disjoint clopen subsets, so the following theorem is immediate:

Theorem 3.11. *A space X is a continuous image of $\mathbb{Q}$ if and only if it is finite ($\neq \emptyset$) or countably infinite.*

Proof. If X is infinite, let x_n be an enumeration of x . We choose a partition of $\mathbb{Q}$ into clopen sets, for example $U_n = (\pi + n, \pi + n + 1)$, with $n \in \mathbb{Z}$, and define $f : \mathbb{Q} \rightarrow X$ so that $f(U_n) = \{x_n\}$.

If X is finite, we do the same with a finite partition of $\mathbb{Q}$ into clopen sets. $\square$

Quotients will not make things so easy for us. There does not seem to be a characterization based on simple properties, but we can give some necessary conditions and some sufficient conditions for a space X to be homeomorphic to a quotient of $\mathbb{Q}$. Separation axioms will not be of much use: there are quotients of $\mathbb{Q}$ that are not T_0 (for example, $\mathbb{N}$ with the trivial topology or the cofinite topology²), and there are countable T_4 spaces that are not quotients of $\mathbb{Q}$ (one of them would be Example 1.6.20 of [En89], which is not sequential, which, as we will see, implies that it is not a quotient of $\mathbb{Q}$).

The only necessary condition for being a quotient of $\mathbb{Q}$ that I have managed to find is a very weak countability axiom, which I study below following part of the article [Fr65].

Definitions 3.12. Given a space X and a subspace A of it, we say that A is sequentially open in X if, given a point a of A and a sequence x_n in X that converges to A , there is a natural number N such that $x_m \in A$ if $m > N$.

We say that a space X is sequential if every sequentially open subset of X is open.

That is, since every open subspace is sequentially open, sequential spaces are those in which the open subspaces are exactly the sequentially open ones. For example, it is straightforward to see that spaces in which every point has a countable neighborhood base (e.g. $\mathbb{Q}$) are sequential. The converse implication, however, is not true: the quotient of $\mathbb{R}$ obtained by collapsing $\mathbb{Z}$ to a single point is sequential (by the proposition we will see next), but it can be checked that its point corresponding to $\mathbb{Z}$ has no countable neighborhood base.

Proposition 3.13. *If $q : X \rightarrow Y$ is a quotient map and X is sequential, Y is sequential.*

Proof. Let A be a sequentially open subset of Y ; let us show that it is open. To do so it suffices to show that $q^{-1}(A)$ is open in X . In turn, for this it suffices to show that $q^{-1}(A)$ is sequentially open in X . But, given $x \in q^{-1}(A)$ and $x_n \in X$ a sequence tending to x , then $q(x_n)$ tends to $q(x) \in A$. Therefore $q(x_n)$ is in A from some N onwards. Therefore x_n will be in $q^{-1}(A)$ for $n > N$, completing the argument. $\square$

²One way to obtain a countable set with the trivial topology as a quotient of $\mathbb{Q}$ is to consider a partition of $\mathbb{Q}$ into infinitely many sets U_n dense in $\mathbb{Q}$. By collapsing each U_n to a point, the quotient we obtain will have the trivial topology. To obtain the cofinite topology, we do something similar, except that now the U_n in the partition will be finite and such that any countable union of them is dense in $\mathbb{Q}$.

Since $\mathbb{Q}$ is sequential, we now have the necessary condition for being a quotient of $\mathbb{Q}$:

Corollary 3.14. *If $\mathbb{Q} \rightarrow Y$ is a quotient map, Y is sequential.* $\square$

In fact, sequentiality is closely related to quotients. A suitable construction will allow us to express any sequential space as a quotient of a metric space:

Proposition 3.15. *Every sequential space is a quotient of a metric space.*

Proof. Let X be sequential. For each $x \in X$ and each sequence $s = \{s_n\}$ in X converging to x , we consider $S(s, x) = \{s_n\}_{n \in \mathbb{N}} \cup \{x\}$, with the topology in which $\{s_n\}$ is open for every n and $s_n \rightarrow x$ (the neighborhoods of x will be the subspaces that contain x and have finite complement). Clearly $S(s, x)$ is metric, since it will be homeomorphic to $\{0\} \cup \{\frac{1}{n}\}_{n \in \mathbb{N}}$ or to a finite discrete space. Now let T be the topological coproduct of all possible $S(s, x)$. T is a coproduct of metric spaces, so it is metrizable.

Each point of T comes from a point of X , which gives rise to a function $f : T \rightarrow X$, which is surjective since for each $x \in X$, the constant sequence x with limit x is in T . f is continuous, since its restriction to each $S(s, x)$ is continuous (to check this it suffices to use the fact that every open set of X containing x contains all but finitely many of the s_n). We will use the fact that X is sequential to show that f is a quotient map. Let $U \subseteq X$ be such that $f^{-1}(U)$ is open in T . Then, given x in U and a sequence $s_n \rightarrow x$, $f^{-1}(U)$ will contain all but finitely many points of $S(s, x)$, so U will contain all the s_n for sufficiently large n . That is, U is sequentially open, so it is open. Therefore f is a quotient map, and we are done. $\square$

This gives us a very simple characterization of sequential spaces.

Theorem 3.16. *The following are equivalent:*

- 1) X is sequential.
- 2) X is a quotient of a space satisfying the 1st axiom of countability.
- 3) X is a quotient of a complete metric space that is locally compact and 0-dimensional.

Proof. The space T in the proof of 3.15 is clearly completely metrizable, locally compact and 0-dimensional, so $1 \Rightarrow 3$. Moreover, $3 \Rightarrow 2$ is obvious and 3.13 proves that $2 \Rightarrow 1$. $\square$

Returning to quotients of the rationals, these will always be sequential, but the converse implication need not hold. In [SW76] an example is given of a countable sequential space that is not a quotient of any metric space of cardinality $< 2^{\aleph_0}$. However, we can find some stronger sufficient conditions for a countable space X to be a quotient of $\mathbb{Q}$:

Proposition 3.17. *If $X \neq \emptyset$ is a finite or countably infinite metric space, it is a quotient of $\mathbb{Q}$.*

Proof. The space $X \times \mathbb{Q}$ is metrizable, countable and without isolated points, and therefore homeomorphic to $\mathbb{Q}$. Moreover, the projection $(x, q) \mapsto x$ from $X \times \mathbb{Q}$ onto X is open, so it is a quotient map. $\square$

A slightly stronger result can be obtained. The following result and its proof are original (as far as I know), although the idea of constructing spaces from sequences is inspired by 1.12 of [Fr65].

Theorem 3.18. *If $X \neq \emptyset$ is finite or countably infinite and satisfies the first axiom of countability, then it is a quotient of $\mathbb{Q}$.*

Proof. By the preceding proposition, it will suffice to prove that X is a quotient of a countable metric space. Let (x_n) be an enumeration (finite or infinite) of X , and for each x_n let $(A_{n,k})_{k=2,3,\dots}$ be a neighborhood base at x_n such that $A_{n,k+1} \subseteq A_{n,k}$.

Our quotient map will be $f : T \rightarrow X$, where T is countable and metric. Let us see how to construct it. We will write T_n for $f^{-1}(x_n)$ for each n , so that the T_n will be nonempty and $T = \bigcup_{n \in \mathbb{N}} T_n$. T will be a subspace of the space of real sequences with finitely many nonzero terms, with metric $d(a, b) = \max\{|b_n - a_n|\}$. The

following paragraph attempts to motivate the construction of T .

We denote the n th unit vector by $e_n = (0, 0, \dots, 0, \overset{n}{1}, 0, \dots)$. Then we will have $e_n \in T_n$, that is, $f(e_n) = x_n$. Now, our quotient map will have to satisfy that the image of any neighborhood of T_n is a neighborhood of x_n , and for this it will suffice that the image of any ball centered at e_n is a neighborhood of x_n . In particular, we can require $f(B(e_n, \frac{1}{k})) = A_{n,k}$. To achieve this, we need that for each point x_m in $A_{n,k}$ there is a point of T_m in $B(e_n, \frac{1}{k})$. This can be achieved by adding to T_m the point $e_n + \frac{e_m}{k+1}$, which lies in $B(e_n, \frac{1}{k})$.

After doing this for all the x_n , we obtain that each set T_n contains the point e_n itself, as well as the points $e_n + \frac{e_m}{k+1}$, whenever x_n belongs to $A_{m,k}$. That is,

$$T_n = \{e_n\} \cup \left\{ e_n + \frac{e_m}{k+1}; m \neq n \text{ and } x_n \in A_{m,k} \right\}.$$

Now, as we said, let $T = \cup_n T_n$ and let the function $f : T \rightarrow X$ be such that $f(T_n) = \{x_n\}$. Since $f(e_n) = x_n$, f is surjective. We will check that f is a quotient map using the following properties:

- (1) All points of T are isolated except possibly the e_n .

Since given another point $p = e_n + \frac{e_m}{k}$, any other distinct point $q = e_a + l e_b$ will satisfy either $n \neq a$, in which case the n th coordinate of q is $\leq \frac{1}{3}$, and therefore $d(p, q) \geq \frac{2}{3}$, or $n = a$ but $m \neq b$, in which case $d(p, q) \geq \frac{1}{k}$, or $n = a, m = b$ but $l \neq \frac{1}{k}$, in which case $d(p, q) \geq \left| \frac{1}{k} - \frac{1}{k+1} \right|$. Therefore for any other point q , $d(p, q) \geq \frac{1}{k(k+1)}$, that is, p is isolated.

- (2) $f(B(e_n, \frac{1}{k})) = A_{n,k}$.

We know by construction that $A_{n,k} \subseteq f(B(e_n, \frac{1}{k}))$. Conversely, if $a \in B(e_n, \frac{1}{k})$, then a will be of the form $e_n + l \cdot e_m$, where l is 0 or $\frac{1}{N+1}$, with $N \geq k$. Therefore either $a = e_n$, and $f(a) \in A_{n,k}$, or $a = e_n + \frac{e_m}{N+1} \in T_m$, in which case $x_m \in A_{n,k} \subseteq A_{n,N}$.

Let us check the continuity of f at each point. At all points except the e_n , f is continuous since by (1) they are isolated. f will be continuous at e_n since if U is a neighborhood of $f(e_n) = x_n$, there will be some $A_{n,k}$ contained in U , so by (2), the neighborhood $B(e_n, \frac{1}{k})$ of e_n will be contained in $f^{-1}(U)$.

Now let $U \subseteq X$ and suppose that $f^{-1}(U)$ is open in T . Then, given $x_n \in U$, $f^{-1}(U)$ contains $B(e_n, \frac{1}{k})$ for some k . By (2), U contains the neighborhood $A_{n,k}$ of x_n . That is, U is open, hence f is a quotient map. $\square$

Note that this last theorem assumes no separation axiom, so it is stronger than the preceding one. In fact, it is another way to prove that a countable set with the trivial topology or the cofinite topology is a quotient of $\mathbb{Q}$, or also that any finite topological space is a quotient of $\mathbb{Q}$.

In the case of $\mathbb{Q}$, we will not devote a section to studying its homogeneity. On the one hand, $\mathbb{Q}$ will be strongly n -homogeneous for every $\mathbb{N}$. This can be proved just as we did for $\mathcal{C}$ in 1.25, using the fact that $\mathbb{Q}$ is 0-dimensional and every clopen subset of $\mathbb{Q}$ is homeomorphic to $\mathbb{Q}$. But $\mathbb{Q}$ is not countable dense homogeneous, since it has countable dense subspaces that are not the whole space. Nevertheless, in Appendix A we see that $\mathbb{Q}$ satisfies another property similar to countable dense homogeneity.

Chapter 4

The real line, $\mathbb{R}$, and the interval $[0,1]$, $\mathbf{I}$

In the first section of the chapter we study the relationship between the order and subspace topologies on subspaces of $\mathbb{R}$, and show that $\mathbb{R}$ is countable dense homogeneous. In the second, we study continua, a class of spaces with interesting connectedness properties that will allow us both to characterize $\mathbf{I} = [0,1]$ and to study the images of $\mathbf{I}$ and $\mathbb{R}$ in the next chapter.

4.1 The order topology on subsets of $\mathbb{R}$

We begin by studying the order topology on subsets of $\mathbb{R}$, given by the usual order on $\mathbb{R}$. In addition to being of interest in its own right, this will be useful when characterizing $\mathbf{I}$. Theorem 4.2 is Proposition 1 of [Fr12].

Definition 4.1. Given a totally ordered set $(X, <)$, its *order topology* is the topology with a base consisting of sets of the form $(a, b)_X = \{x \in X; a < x < b\}$, $(a, \infty)_X = \{x \in X; a < x\}$ and $(-\infty, b)_X = \{x \in X; x < b\}$ ¹.

Since this topology is defined in terms of the order, any order isomorphism (an increasing bijective function) between two linearly ordered sets will be a homeomorphism between their order topologies. We will also use in 4.15 the fact that these topologies are T_2 . This holds since, given $x, y \in X$ with $x < y$, if there is an element c in $(x, y)_X$, then $(-\infty, c)$ and (c, ∞) separate x from y . If, on the other hand, no such c exists, then $(-\infty, y)$ and (x, ∞) separate x and y . In fact, although we will not prove it here, all order topologies are normal.

On many subsets of $\mathbb{R}$, the order topology coincides with the subspace topology. Examples of these spaces are, as we will see, $\mathbb{R}$, $\mathbf{I}$, $\mathbb{Q}$, $\mathbb{I}$ and $\mathcal{C}$. However, on other occasions the two topologies do not coincide. For example, if we consider $A = \{0\} \cup (1, 2)$, in its subspace topology 0 is an isolated point, whereas in the order topology it is not. The following result shows that this is because A has a ‘gap’, $(0, 1]$, which is a half-closed interval.

Theorem 4.2. Let A be a subset of $\mathbb{R}$. The topology of A as a subset of $\mathbb{R}$, $\mathcal{T}_{A, \mathbb{R}}$, will always be finer than the order topology of A , $\mathcal{T}_{A, <}$. Moreover, the two topologies coincide if and only if $\mathbb{R} - A$ has no connected component of the form $(a, b]$ or $[b, a)$, with $a, b \in \mathbb{R}$.

Proof. The first statement follows from the fact that for each $a, b \in A$, $(a, b)_A = (a, b)_{\mathbb{R}} \cap A$, $(\infty, b)_A = (\infty, b)_{\mathbb{R}} \cap A$ and $(a, \infty)_A = (a, \infty)_{\mathbb{R}} \cap A$, so the open sets of $\mathcal{T}_{A, <}$ are also open in $\mathcal{T}_{A, \mathbb{R}}$.

If A has a gap of the form $(a, b]$, then $a \in A$ and there is a decreasing sequence b_n tending to b , with $b_n \in A$ (if these conditions did not hold, we would be contradicting the maximality of $(a, b]$ as a connected component of $\mathbb{R} - A$). Therefore in $\mathcal{T}_{A, <}$, every neighborhood of a contains infinitely many b_n , whereas in $\mathcal{T}_{A, \mathbb{R}}$ this is not

¹Normally we would have to specify the order as well as the set when naming the intervals, but since here we will only deal with subsets of $\mathbb{R}$ with their usual order, there is no risk of confusion.

the case. Therefore $\mathcal{T}_{A,\mathbb{R}} \neq \mathcal{T}_{A,<}$. The same happens with gaps of the form $[b, a)$, of course.

Suppose that A has no gaps of the form $(a, b]$ or $[b, a)$. To show that $\mathcal{T}_{A,\mathbb{R}} \subseteq \mathcal{T}_{A,<}$ it will suffice to show that for all $a, b \in \mathbb{R}$, $(-\infty, b) \cap A$ and $(a, \infty) \cap A$ are open in $\mathcal{T}_{A,<}$. Let us consider the case $(-\infty, b)$; the other is symmetric. If $b \in A$, this is immediate since $(-\infty, b) \cap A = (-\infty, b)_A$, which is open in $\mathcal{T}_{A,<}$. Otherwise, we denote by B the connected component of b in $\mathbb{R} - A$. We divide into cases according to the type of interval B is:

- $B = [c, d]$, with $c \leq b \leq d$. Then there is a sequence $c_n \rightarrow c$ in A , and $(-\infty, b) \cap A = \bigcup_{i \in \mathbb{Z}^+} (-\infty, c_n)_A$ is open.
- $B = (-\infty, d]$, with $d \leq b$. Then $(-\infty, b) \cap A = \emptyset$ is open in A .
- $B = [c, \infty)$, with $c \leq b$. In this case $(-\infty, b) \cap A = A$ is open in A .
- $B = (c, d)$, with $c < b$ or $c = -\infty$ and $d > b$. Then $(-\infty, b) \cap A = (-\infty, d)_A$ is open in A .
- $B = (c, \infty)$, with $c < b$ or $c = -\infty$. Then $(-\infty, b) \cap A = A$ is open in A .

□

We move on to prove that $\mathbb{R}$ has the always surprising property (even though almost all the spaces we are studying have it) of countable dense homogeneity. In fact, we will not generalize here, but in [Br13] it is proved that $\mathbb{R}^n$ is countable dense homogeneous for every n . To do this we use a well-known lemma about orders, which will also be useful when characterizing **I**. Recall that a linear order on a set is dense if, given any x, y with $x < y$, there always exists z with $x < z < y$.

Lemma 4.3 (Cantor). *Given countable sets X, Y with dense linear orders without greatest or least elements, there is an order isomorphism $f : X \rightarrow Y$.*

Proof. Let $(x_n), (y_n)$ be enumerations of X and Y . We will define f recursively.

- Step 1: We define $f(x_1) = y_1$.
- Step n : Suppose that we have already defined $f(x_1), \dots, f(x_{n-1}), f^{-1}(y_1), \dots, f^{-1}(y_{n-1})$, and that f is strictly increasing on its domain. We denote by $a_1, \dots, a_k$ the elements of X at which we have already defined f , with $a_1 < \dots < a_k$. Therefore $f(a_1) < \dots < f(a_k)$. To define $f(x_n)$ we have several cases:
 - $x_n < a_1$. Then we define $f(x_n)$ as an element of Y less than $f(a_1)$ (one exists since Y has no least element).
 - $a_i < x_n < a_{i+1}$ for some i . Then we define $f(x_n)$ so that $f(a_i) < f(x_n) < f(a_{i+1})$ (one exists since Y is dense).
 - $x_n > a_k$. We define $f(x_n)$ as an element of Y greater than $f(a_k)$.

By the way we have defined $f(x_n)$ it is clear that f remains injective and increasing on its domain.

Similarly, using the fact that the order on X is dense and has no greatest or least element, we can find $f^{-1}(y_n)$ (if it was not already defined) so that f remains increasing.

After repeating this step for each natural number n , it is clear that the f we obtain will be bijective (at each step we have defined $f(x_n)$ to be distinct from the other elements of the image and have defined $f^{-1}(y_n)$). It is also obvious from its construction that it will be increasing. □

Proposition 4.4. *$\mathbb{R}$ is countable dense homogeneous.*

Proof. Let D, E be two countable dense subsets of $\mathbb{R}$. By 4.3 there will be an increasing bijective function $g : D \rightarrow E$. We will obtain an extension $f : \mathbb{R} \rightarrow \mathbb{R}$ of it that is a homeomorphism. To do so, given $x \in \mathbb{R}$, we will define $f(x) = \sup\{g(q); q \in D, q < x\}$. f is well-defined since, given $r \in D$ that is $> x$, $g(r)$ will be an upper bound for $\{g(q); q \in D, q < x\}$.

If $q \in D$, let us show that $f(q) = g(q)$. Given $p < q$ in D , we will have $g(p) < g(q)$. Since this holds for every p , $f(q) \leq g(q)$. Moreover, since E is dense, there is a strictly increasing sequence e_n in E tending to $g(q)$. Therefore, since $f(q) > g(g^{-1}(e_n)) = e_n$ for every n , $f(q) \geq g(q)$. Hence $f(q) = g(q)$.

Moreover, given $x < y$ in $\mathbb{R}$, there are two elements c, d in D with $x < c < d < y$, and it follows from the definition of f that $f(x) \leq g(c) < g(d) \leq f(y)$. This shows that f is strictly increasing, and therefore injective. If we prove that f is surjective, we will have shown that f is an order isomorphism, and therefore a homeomorphism.

Let us show that f is surjective: given $y \in \mathbb{R}$, we define $x := \sup\{q \in D; g(q) < y\}$. Then $f(x) \leq y$. Moreover, for any $r \in E$ less than y , we will have $g^{-1}(r) \leq x$, so $r \leq f(x)$. Taking a sequence r_n in E tending to y from below, we see that $y \leq f(x)$. Therefore $y = f(x)$. $\square$

In Appendix A there is an even stronger homogeneity result for $\mathbb{R}$.

4.2 Continua, cut points. Characterization of **I**

A *continuum* is a T_2 , compact and connected space. A subcontinuum of a space is a subspace of it that is a continuum. Examples of continua are the cubes $\mathbf{I}^n$, the spheres $\mathbb{S}^n$ with $n \geq 1$, or any other compact connected subspace of $\mathbb{R}^n$. There are also examples beyond $\mathbb{R}^n$, of course: any product of continua is a continuum. For example, $\mathbf{I}^\alpha$ is a continuum for any cardinal α (and for $\alpha > \aleph_0$, it is not even metrizable). Our aim will be to characterize **I** as a continuum. This section will closely follow Chapter 28 of [Wi70], fixing along the way a detail in the proof of 28.7 (here, Proposition 4.10).

Recall that a directed set $(X, \preceq)$ is a set X with a relation $\preceq$ on X that is reflexive, transitive, and such that for every two elements x, y in X there is some z with $x \preceq z, y \preceq z$. For example, linearly ordered sets are directed. A family $\mathcal{K}$ of sets is directed by reverse inclusion if, given finitely many sets in $\mathcal{K}$, there is another contained in their intersection.

Proposition 4.5. *Let X be T_2 , and let $\mathcal{K} = \{K_\alpha\}_{\alpha \in A}$ be a collection of subcontinua of X directed by reverse inclusion. Then $K = \bigcap_{\alpha \in A} K_\alpha$ is a continuum. Moreover, if $K_\alpha \neq \emptyset$ for every α , $K \neq \emptyset$.*

Proof. We may assume that X is compact: otherwise, we replace X by K_α for some nonempty K_α , and K_β by $K_\alpha \cap K_\beta$.

That K is compact is obvious. Suppose that it is not connected, that is, there is a separation of K , $K = X_1 \cup X_2$, with X_1 and X_2 disjoint and closed in K and $x \in X_1$. Then X_1 and X_2 are closed in X , so since X is normal, there are disjoint open sets U, V in X with $X_1 \subseteq U, X_2 \subseteq V$. $U \cup V$ is a neighborhood of K . $X - K_\alpha$ is an open cover of the compact set $X - (U \cup V)$. Therefore there will be finitely many $K_{\alpha_1}, \dots, K_{\alpha_n}$ whose complements cover $X - (U \cup V)$. Writing K_{α_0} for an element of $\mathcal{K}$ contained in all of them, K_{α_0} is contained in $U \cup V$. This is a contradiction, since then U and V form a separation of K_{α_0} , contradicting its connectedness.

If K_α is nonempty for every α , then, since $\mathcal{K}$ is directed by reverse inclusion, it will be a collection of closed sets with the finite intersection property, so its intersection, K , is nonempty. $\square$

Definition 4.6. A continuum K is irreducible with respect to a subset A if no proper subcontinuum of K contains A . If $A = \{a, b\}$, we say that K is irreducible between a and b .

Proposition 4.7. *Given a continuum K and a subspace A of it, there is a subcontinuum of K irreducible with respect to A .*

Proof. The set $\mathcal{K}$ of subcontinua of X that contain A is partially ordered by inclusion. Moreover, by 4.5 every chain in $\mathcal{K}$ has a minimal element (the intersection of its elements). By Zorn's lemma, there will be a minimal element of $\mathcal{K}$, which will therefore be irreducible with respect to A . $\square$

For example, any arc between 2 points in $\mathbb{R}^n$ is a subcontinuum irreducible between the two points.

Definition 4.8. Let X be connected and T_1 . We say that $p \in X$ is a cut point of X if $X - \{p\}$ is not connected.

A cut of X is a triple (p, U, V) , where p is a cut point of X and U, V are two disjoint nonempty open sets whose union is $X - \{p\}$.

Proposition 4.9. *If (p, U, V) is a cut of a continuum K , then $U \cup \{p\}$ and $V \cup \{p\}$ are connected (and therefore continua).*

Proof. $U \cup \{p\}$ and $V \cup \{p\}$ are closed, and therefore compact, being the complements of V and U in K . To show that they are connected, we consider the function

$$\begin{aligned} f: K &\rightarrow K; \\ x &\mapsto \begin{cases} x & \text{if } x \in U \cup \{p\} \\ p & \text{if } x \in V \cup \{p\} \end{cases} \end{aligned}$$

Since f is continuous on $U \cup \{p\}$ and on $V \cup \{p\}$, it is continuous by the pasting lemma. Therefore $U \cup \{p\}$ is the continuous image of K , which is connected, so it is connected. Similarly, we can show that $V \cup \{p\}$ is connected. $\square$

Proposition 4.10. *If (p, U, V) is a cut of a continuum K , then U and V each contain a non-cut point of K .*

Proof. Suppose that all points x of U are cut points of K , giving a cut (x, U_x, V_x) , and we will reach a contradiction. If $x \in U$, U_x and V_x cannot both intersect $V \cup \{p\}$, since they would separate it, and by the preceding proposition $V \cup \{p\}$ is connected. Therefore one of the open sets, which will be the one we call U_x , is contained in U (and V is contained in V_x).

Something I will use a couple of times is that if x, y are points of U and $x \in U_y$, then $U_x \subseteq U_y$. This is because otherwise U_x and V_x would both intersect $V_y \cup \{y\}$, disconnecting it.

Then $\mathcal{U} = \{U_x \cup \{x\}\}_{x \in U}$ is a family of continua contained in $U \cup \{p\}$. Moreover, $\mathcal{U}$ is partially ordered by reverse inclusion. Let us show that $\mathcal{U}$ has maximal elements using Zorn's lemma. Given a chain $\mathcal{A} = \{U_x \cup \{x\}\}_{x \in \mathcal{A}}$, since it is a directed collection of nonempty continua, its intersection will be, by 4.5, a nonempty continuum, L . Let q be a point of L . Then $q \in U_x \cup \{x\}$ for every $x \in \mathcal{A}$, so $U_q \subseteq U_x$ for every $x \in \mathcal{A}$, hence $U_q \cup \{q\} \subseteq L$. That is, $U_q \cup \{q\}$ is an upper bound for the chain. By Zorn's lemma, we will have a point r such that $U_r \cup \{r\}$ is maximal with respect to reverse inclusion in $\mathcal{U}$. But, taking a point $s \in U_r$, we again have $U_s \subseteq U_r$. Therefore $U_s \cup \{s\} \subsetneq U_r \cup \{r\}$, contradicting the maximality of $U_r \cup \{r\}$. $\square$

Theorem 4.11. *Every continuum K with more than one point has at least two non-cut points.*

Proof. If K has cut points, the result is true by 4.10. Otherwise, the result is obvious. $\square$

This result has much to do with **I**: the characterization we will give of **I** is that it is the only metrizable continuum with exactly 2 cut points. But we still need more results for this. The first tells us that no proper subcontinuum of a continuum K can contain all its non-cut points.

Theorem 4.12. *A continuum K is irreducible with respect to its set of non-cut points.*

Proof. Suppose that there is a subcontinuum $L \subseteq K$ that contains all the non-cut points of K . Given $x \in K - L$, there will be a cut (x, U, V) of K . Since L is connected, it cannot intersect both U and V , so we assume, for example, $L \subseteq U$. But by 4.10, there are non-cut points of K in V , which contradicts the fact that L contains all the non-cut points of K . $\square$

We move on to an order relation that we can establish between cut points.

Definition 4.13. A cut point p in a space X separates a from b if there is a cut of X , (p, U, V) , with $a \in U$ and $b \in V$.

We denote by $E(a, b)$ the set consisting of a, b and all the cut points that separate them.

We define the relation $\leq$ on $E(a, b)$ by: $a \leq p \leq b$ for every p , and for the remaining points, $p_1 \leq p_2$ iff $p_1 = p_2$ or p_1 separates a from p_2 .

Proposition 4.14. *If a, b are points in a continuum K , the relation $\leq$ from 4.13 is a total order on $E(a, b)$.*

Proof. It suffices to show that $\leq$ is a linear order on $E(a,b) - \{a,b\}$. Given $p \in E(a,b) - \{a,b\}$, we denote by (p, U_p, V_p) a separation of a and b by p , with $a \in U_p$ and $b \in V_p$.

Given distinct p and q in $E(a,b) - \{a,b\}$, U_p and U_q intersect at a , and V_p and V_q intersect at b . We distinguish two possible cases.

- If $q \in V_p$, p separates a from q , $p \leq q$. V_q cannot intersect $U_p \cup \{p\}$, since then U_q and V_q would separate $U_p \cup \{p\}$. Therefore $V_q \subseteq V_p$.
- If $q \in U_p$, U_q cannot intersect $V_p \cup \{p\}$, since then U_q and V_q would separate $V_p \cup \{p\}$. Therefore $U_q \subseteq U_p$. Taking complements, $V_p \cup \{p\} \subseteq V_q \cup \{q\}$. Since $p \neq q$, we have $p \in V_q$. That is, q separates a from p , $q \leq p$.

The relation $\leq$ is obviously reflexive. As we have seen in the division into cases, given two elements of $E(a,b) - \{a,b\}$, p and q , we have $p \leq q$ or $q \leq p$. It remains to check antisymmetry and transitivity.

If $p \leq q$ and $q \leq p$, $q \in V_p$ and $p \in V_q$ for some cuts (p, U_p, V_p) and (q, U_q, V_q) . By what we saw in the first case, we will have $V_q \subseteq V_p$ and $V_p \subseteq V_q$, hence $V_p = V_q$. Therefore the boundary of V_p is equal to that of V_q , that is, $p = q$.

If $p \leq q$ and $q \leq r$, $q \in V_p$ and $r \in V_q$ for some cuts (p, U_p, V_p) and (q, U_q, V_q) . Therefore $V_q \subseteq V_p$ and $V_r \subseteq V_q$, hence $V_r \subseteq V_p$. Therefore $r \in \overline{V_r} \subseteq \overline{V_p} = V_p \cup \{p\}$. Since $r \neq p$, $r \in V_p$, that is, $p \leq r$. $\square$

Now that we know that $\leq$ is a linear order on $E(a,b)$, it is natural to consider its order topology. We will see a case in which the order topology on $E(a,b)$ and the topology inherited as a subset of K coincide.

Proposition 4.15. *If K is a continuum with only two non-cut points, a and b , then $E(a,b) = K$ and the topology on K is the order topology.*

Proof. $E(a,b) = K$ since, given a point $p \in K - \{a,b\}$, it has an associated cut (p, U_p, V_p) . By 4.10, U_p and V_p each contain a non-cut point. These points can only be a and b , so p separates a and b . We move on to compare the topology of K with the order topology.

In the division into cases in 4.14 we checked that, given $p, q \in K - \{a,b\}$, $p < q$ iff $q \in V_p$ iff $p \in U_q$. Therefore a subbase for the order topology will consist of the subsets of the form $(-\infty, q) = U_q$ and $(p, \infty) = V_p$. These are open in the topology of K .

Now we consider the identity function on K , 1_K , where the domain has the topology of K and the codomain has the order topology. The domain of the function is compact, and the image is T_2 . Moreover, by the preceding paragraph the function is continuous. Being bijective, f is a homeomorphism. $\square$

We are now ready for the characterization of **I** as a continuum.

Theorem 4.16. *If K is a metrizable continuum and has exactly 2 non-cut points, it is homeomorphic to **I**.*

Proof. Since K and **I** have the order topology by 4.15 and 4.2, it suffices to prove that there is an order isomorphism (an increasing bijective function) between them.

Since K is metrizable and compact, it has a countable dense subset D , which we may assume does not contain the non-cut points, a and b . This subset D has no least or greatest element, since a, b are in its closure. Moreover, given two points r, s in D , $(r, s)_K$ is nonempty since K is connected, so since D is dense in K , $(r, s)_D$ is nonempty. That is, the order on D is dense. By Lemma 4.3, there will be an order isomorphism g from D onto $\mathbb{Q} \cap (0, 1)$.

It will then suffice to extend this isomorphism g to an order isomorphism $f : K \rightarrow [0, 1]$. Naturally, we define $f(a) = 0$, $f(b) = 1$. If p is another point of K , we define $f(p)$ as $\sup\{g(q); q \leq p\}$. In this way, it is clear that if q is a point of D , then if $p \leq q$ we will have $f(p) \leq g(q)$, and if $p > q$, $f(p) > g(q)$. Since there are points of D between every 2 distinct points of K , this implies that f is injective. Moreover, f is obviously increasing.

It remains to show that f is surjective. But given a point $x \in \mathbf{I}$, there is a sequence q_n of rational numbers tending to x from below. Taking $p = \sup\{g^{-1}(q_n)\}$, it is obvious that $f(p) \leq x$. Moreover, $f(p) \geq x$ since for any rational q that is $\leq x$, $f(p) \geq q$, since there is some n with $q_n > q$. Thus $f(p) = x$. $\square$

One may wonder about the homogeneity of $\mathbf{I}$. This is the only one of the spaces we will study that is not homogeneous, since all its points are cut points except 0 and 1. This implies that the space is not countable dense homogeneous, since some countable dense subsets of $\mathbf{I}$ contain $\{0, 1\}$ and others do not. However, if we require that 0 and 1 do not belong to the countable dense subsets, we can obtain a result:

Proposition 4.17. *Let A, B be countable dense subsets of $\mathbf{I}$ that contain neither 0 nor 1. Then there is a homeomorphism $f : \mathbf{I} \rightarrow \mathbf{I}$ with $f(A) = B$.*

Proof. Since $(0, 1)$ is homeomorphic to $\mathbb{R}$, which is countable dense homogeneous, there will be a homeomorphism $f : (0, 1) \rightarrow (0, 1)$ with $f(A) = B$. If we wish, we can take f to be increasing, as in the proof of 4.4. Then, since f is an increasing bijection from $(0, 1)$ to $(0, 1)$, we will have $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{x \rightarrow 1} f(x) = 1$. Therefore, defining $f(0) = 0$ and $f(1) = 1$, we have a continuous bijective function $f : \mathbf{I} \rightarrow \mathbf{I}$ from a compact space to a T_2 space, that is, a homeomorphism. $\square$

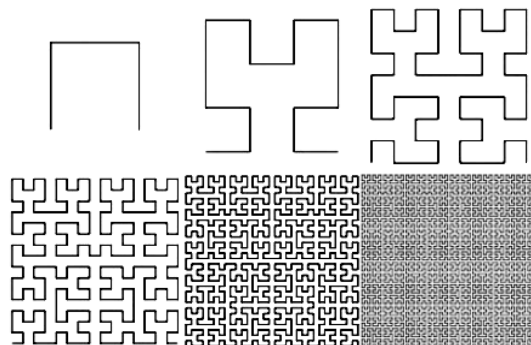
Chapter 5

Peano continua

We move on to study the continuous images of $\mathbf{I}$ and $\mathbb{R}$. The famous Hahn-Mazurkiewicz theorem shows that the T_2 continuous images of $\mathbf{I}$ are exactly the Peano continua, which we describe below. For the images of $\mathbb{R}$ there does not seem to be such a nice characterization, but in the second section we give a class of them broader than the Peano spaces. Finally, we use these results to give a characterization of $\mathbb{R}$ in the third section.

5.1 Peano continua. The Hahn-Mazurkiewicz Theorem

Recall that a curve or path in a space X is a continuous function $f : \mathbf{I} \rightarrow X$. A similar but more restrictive concept is that of an arc: $f : \mathbf{I} \rightarrow X$ is an *arc* from a to b if f is a homeomorphism onto its image, $f(0) = a$ and $f(1) = b$. In general, we will call spaces homeomorphic to $\mathbf{I}$ arcs. The question we ask in this section is, which spaces are continuous images of $\mathbf{I}$? In the late nineteenth century, Peano discovered that there are curves that cover the square, $\mathbf{I}^2$. He constructed an example as the uniform limit of a sequence of curves. The figure shows a similar construction by Hilbert. By constructing similar curves, one can see that $\mathbf{I}^n$ is an image of $\mathbf{I}$ for every n .



In this section we will characterize which T_2 spaces are continuous images of $\mathbf{I}$. These spaces will obviously be compact and connected. Moreover, by 1.15, they will be metrizable. Another property they will have, which is not so obvious, is local connectedness (a space is locally connected if every point has a base of open connected neighborhoods). To prove this we will use the following technical lemma.

Lemma 5.1. *If $f : X \rightarrow Y$ is a quotient map and X is locally connected, Y is locally connected.*

Proof. We will use the characterization that Y is locally connected iff for every open set V of Y , the connected components of V are open (27.9 of [Wi70]).

Let V be an open set of Y , let y be a point of V and $C(y)$ the connected component of y in V . We want to show that $C(y)$ is open. Since f is a quotient map, it will suffice that $C = f^{-1}(C(y))$ is open. Given $x \in C$, since X is locally connected, the connected component U_x of x in $f^{-1}(V)$ will be open. Since $f(U_x) \subseteq V$ is connected, $f(U_x) \subseteq C(y)$, hence U_x is contained in C . Therefore x is an interior point of C , that is, C is open. $\square$

If X is a T_2 space and $f : \mathbf{I} \rightarrow X$ is continuous and surjective, f is closed and therefore a quotient map, so by the preceding lemma X is locally connected.

Definition 5.2. A Peano continuum is a metrizable, compact, connected and locally connected space.

From what we have seen, the continuous images of $\mathbf{I}$ that are T_2 are Peano continua. Throughout the rest of this section we will prove the converse implication, closely following Chapter 31 of [Wi70]. A first step towards this is to prove that Peano continua are path connected. In fact, we will prove something stronger: that they are arc connected. We will thus obtain that a T_2 space is path connected if and only if it is arc connected, an intuitive result but by no means easy to prove. We begin with some lemmas on connectedness and chains of open sets.

Definition 5.3. A chain between a and b in a space X is a sequence $(U_1, \dots, U_n)$ of open sets in X with $a \in U_1$, $b \in U_n$ and $U_i \cap U_{i+1} \neq \emptyset \forall i$. We say that the chain is simple if a belongs only to U_1 , b belongs only to U_n and $U_i \cap U_j = \emptyset \iff |i - j| > 1$.

In the following theorem we will use the easily checked fact that the union of the elements of a chain of connected sets is connected. This can be checked since, if we have the chain of connected sets $(U_1, \dots, U_n)$, then any clopen set containing a point of U_1 will have to contain all of U_1 since U_1 is connected. Therefore the clopen set intersects U_2 , and therefore contains U_2 , and thus by induction we see that the clopen set contains all the elements of the chain.

Lemma 5.4. If X is connected and $\mathcal{U}$ is an open cover of X , given a and b in X there will be a simple chain of open sets from $\mathcal{U}$ between a and b .

Proof. We consider Z , the set of points of X connected to a by some simple chain of open sets from $\mathcal{U}$. This set is clearly open, since if $x \in Z$ and the simple chain $U_1, \dots, U_n$ connects a and x , then the neighborhood U_n of x is contained in Z . Let us show that Z is closed. Let $x \in \overline{Z}$. Let $U \in \mathcal{U}$ be a neighborhood of x , and let $y \in U \cap Z$ and $(U_1, \dots, U_n)$ be a chain between a and y . We have two cases: if x belongs to some U_i , we take the smallest i for which this holds, and then $(U_1, \dots, U_i)$ is a simple chain from a to x , hence $x \in Z$. If x belongs to no U_i , let i be the smallest index such that $U_i \cap U \neq \emptyset$. Then $(U_1, \dots, U_i, U)$ is a simple chain from a to x , so $x \in Z$. $\square$

The following theorem is crucial, and its proof is quite geometric and cumbersome to formalize. So much so that in many books the proof is incorrect or at least incomplete, as explained in [Ba84]. The proof here is a modification of the one in [Wi70].

Theorem 5.5. Every Peano continuum is arc connected.

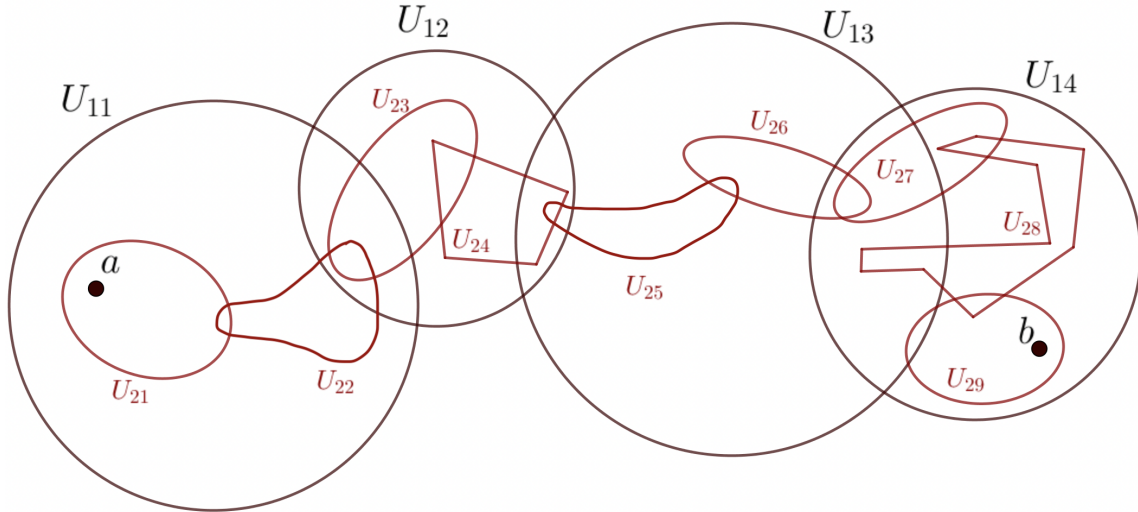
Proof. Let X be our space, and let a and b be two of its points. By the preceding lemma there is a simple chain from a to b of connected open sets of diameter < 1 , which we will call $\mathfrak{C}_1 = (U_{1,1}, \dots, U_{1,n_1})$.

To construct an arc from a to b we will use a sequence of simple chains from a to b constructed recursively. We will call them $\mathfrak{C}_k = (U_{k,1}, \dots, U_{k,n_k})$, for $k \geq 2$, and they will satisfy these properties:

- (1) $\mathfrak{C}_k$ is a simple chain from a to b of connected open sets of diameter $< \frac{1}{2^k}$.
- (2) All the open sets $U_{k,i}$ of $\mathfrak{C}_k$ satisfy $\overline{U_{k,i}} \subseteq U_{k-1,j}$ for some j .
- (3) For each k , there are indices $0 = j_0 < j_1, \dots, j_{n_k} = n_{k+1}$ such that, in the chain $\mathfrak{C}_{k+1}$, the $U_{k+1,l}$ with $j_{i-1} < l \leq j_i$ are contained in $U_{k,i}$.

To construct $\mathfrak{C}_{k+1}$ from $\mathfrak{C}_k$, we begin by finding for each point p of $U_{k,i}$ an open connected neighborhood of p , V_p , with $\overline{V_p} \subseteq U_{k,i}$ and with diameter $< \frac{1}{2^{k+1}}$. This gives open covers of the $U_{k,i}$.

Now, for each $i = 1, \dots, n_k - 1$, we choose an $x_i \in U_{k,i} \cap U_{k,i+1}$, and write $x_0 = a, x_{n_k} = b$. For each $i = 1, \dots, n_k$, we find a simple chain $\mathfrak{D}_i$ inside $U_{k,i}$ from x_{i-1} to x_i , formed by the V_p defined above. Joining all these chains to form a new chain, $(V_1, \dots, V_m)$, this will be a chain from a to b , but it need not be simple: there may be unwanted intersections. However, this can easily be fixed: if V_i intersects V_j at some point, with $|j - i| \geq 2$, then we can omit all the V_k with k between i and j . Since in this process we are removing elements of the chain, we can only repeat it finitely many times, and when it can no longer be repeated we will have obtained a simple chain, which is $\mathfrak{C}_{k+1}$.


 Figure 5.1: Example of the construction of the chains $\mathfrak{C}_1$ and $\mathfrak{C}_2$.

To check that this chain satisfies (3), we take $j_0 = 0$ and the remaining j_i so that U_{k+1,j_i} is the last element of $\mathfrak{D}_i$ that remains in $\mathfrak{C}_{k+1}$. Such an element must exist: otherwise, every element of $\mathfrak{C}_{k+1}$ would be contained in an element of $\mathfrak{C}_k$ other than $U_{k,i+1}$. Therefore the unions $\bigcup_{j=0}^i U_{k,j}$ and $\bigcup_{j=i+2}^{n_k} U_{k,j}$ would be open sets disconnecting the union of the elements of $\mathfrak{C}_{k+1}$. This is a contradiction, since the union of the elements of a chain of connected sets must be connected.

We now proceed to construct the arc from a to b . For each k , we consider C_k , the union of the closures of all the elements of $\mathfrak{C}_k$. This set is clearly compact and connected (the closure of a connected set is connected and the union of a chain of connected sets is connected), so it is a continuum. Thus by 4.5, $C = \bigcap_{k=1}^{\infty} C_k$ is also a continuum. If we check that every point of C except a and b is a cut point, we will have that C is an arc from a to b by 4.11 and 4.16, and we will be done.

Let $x \in C - \{a, b\}$. By property (2), it is clear that x belongs to some element of $\mathfrak{C}_k$ for each k . In fact, it will belong to 1 or 2 elements of each $\mathfrak{C}_k$, since the $\mathfrak{C}_k$ are simple chains. We denote by $n_{k,1}$ and $n_{k,2}$, with $n_{k,1} \leq n_{k,2}$, the indices (consecutive or equal) such that x belongs to $U_{k,n_{k,1}}$ and $U_{k,n_{k,2}}$. Since x is neither a nor b , it is clear that for sufficiently large k we will have $n_{k,1} > 0$ and $n_{k,2} < n_k$. For each k we define the following partition of C_k :

$$L_k = \overline{U_{k,n_{k,1}}} \cup \overline{U_{k,n_{k,2}}}.$$

$$A_k = \left(\bigcup_{i=0}^{n_{k,1}-1} U_{k,i} \right) - L_k.$$

$$B_k = \left(\bigcup_{i=n_{k,2}+1}^{n_k} U_{k,i} \right) - L_k.$$

That is, L_k will be the closure of the union of the elements of $\mathfrak{C}_k$ that contain x , A_k will be the ‘part of the chain $\mathfrak{C}_k$ before L_k ’ and B_k will be the ‘part of the chain $\mathfrak{C}_k$ after L_k ’. Moreover, A_k and B_k are disjoint open sets since $\mathfrak{C}_k$ is a simple chain, and the diameter of L_k tends to 0 as k tends to ∞ . It also follows from (2) that $L_{k+1} \subseteq L_k$ for every k . Let us show that x is a cut point of C by giving a separation of $C - \{x\}$ into the two open sets:

$$A = \bigcup_{k=1}^{\infty} A_k \cap C \quad \text{and} \quad B = \bigcup_{k=1}^{\infty} B_k \cap C,$$

It is easy to see that $A \cup B = C - \{x\}$, since, as $\text{diam}(L_k) \rightarrow 0$, any point of C except x will have to be in A_k or B_k for sufficiently large k . We also have to show that A and B are disjoint. If they were not, there would be k and l with $A_k \cap B_l \cap C \neq \emptyset$. Since A_k and B_k are disjoint, $k \neq l$. Suppose that $k < l$ (the other case is symmetric). It suffices to show that if a point y of C belongs to A_k , then it belongs to A_{k+1} , since

then by induction it will belong to A_l and therefore cannot belong to B_l , contradicting $A_k \cap B_l \cap C \neq \emptyset$.

Suppose, then, that we have a point $y \in A_k \cap C$. Let us see where y lies in the chain $\mathfrak{C}_{k+1}$, using the notation of (3). Since it belongs to A_k , y does not belong to $U_{k,o}$ for any $o \geq n_{k,1}$. Thus by (3), in $\mathfrak{C}_{k+1}$, y will not belong to elements $U_{k+1,\alpha}$ with $\alpha > j_{n_{k,1}-1}$. On the other hand, x cannot belong to any $U_{k+1,\beta}$ with $\beta \leq j_{n_{k,1}-1}$, since then x would belong to some $U_{k,N}$ with $N \leq n_{k,1} - 1$, and this does not happen by the definition of $n_{k,1}$. Thus x will appear in elements $U_{k+1,\beta}$ with $\beta > j_{n_{k,1}-1}$.

Therefore, since the indices β at which x appears are greater than the indices α at which y appears in the chain $\mathfrak{C}_{k+1}$, y will not belong to B_{k+1} . Nor will it belong to L_{k+1} since $L_{k+1} \subseteq L_k$, so $y \in A_{k+1}$, as desired. $\square$

Corollary 5.6. *If A is a connected open subspace of a Peano continuum X , then A is arc connected.*

Proof. In the preceding proof, if the two points a and b belong to A , using the fact that A is open and connected in X (and therefore locally connected) and 5.4, we can form the chain $\mathfrak{C}_1$ so that its elements are contained in A . Since the arc that is formed is contained in the union of the elements of $\mathfrak{C}_1$, the arc we obtain in the proof will lie in A , that is, A is arc connected. $\square$

Theorem 5.7. *A T_2 space is path connected if and only if it is arc connected.*

Proof. If X is arc connected, it is obvious that it will be path connected. If X is path connected, given two points a and b , there is a continuous $f : \mathbf{I} \rightarrow X$ with $f(0) = a$, $f(1) = b$. But since, by what we saw at the beginning of the section, $f(\mathbf{I})$ is a Peano continuum, $f(\mathbf{I})$ will be arc connected. That is, there is an arc joining a and b in $f(\mathbf{I})$, and therefore in X . $\square$

The T_2 condition is necessary for the preceding theorem: the line with two origins, which is T_1 but not T_2 , is path connected, but it is not difficult to check that the two origins cannot be connected by an arc.

The strategy we will use to show that Peano continua are images of $\mathbf{I}$ begins by observing that, by the Hausdorff-Alexandroff theorem 1.12, there is a continuous function onto X from the Cantor set, $g : \mathcal{C} \rightarrow X$. We will then have to extend g to a continuous function $f : \mathbf{I} \rightarrow X$, that is, we have to define an image under f for each ‘gap’ (a_i, b_i) of the Cantor set. Since we now know that X is path connected, we can define f on $[a_i, b_i]$ as a path from $g(a_i)$ to $g(b_i)$, and in fact that is the idea we will use. However, since we have to define the images on infinitely many intervals, the pasting lemma does not apply and the resulting f may fail to be continuous. To overcome this difficulty we will develop the concept of uniform local path connectedness.

Let us briefly recall the concept of a Lebesgue number of a cover: given a compact metric space X and a cover $\mathcal{U}$ of it, the function

$$\begin{aligned} f : X &\rightarrow \mathbb{R}; \\ x &\mapsto \sup\{r > 0; B(x, r) \subseteq U \text{ for some } U \in \mathcal{U}\} \end{aligned}$$

is positive and continuous (it has Lipschitz constant 1), so it attains a minimum $\delta > 0$. We call this δ a Lebesgue number of the cover, and by its definition any subset of X of diameter $< \delta$ will be contained in some element of $\mathcal{U}$.

Lemma 5.8. *A compact locally connected metric space is uniformly locally connected, that is, for each $\varepsilon > 0$ there is some $\delta > 0$ such that if $d(x, y) < \delta$, x and y lie in a connected subspace of X of diameter $< \varepsilon$. Moreover, the subspace can be chosen to be open.*

Proof. If we start with a cover $\mathcal{U}$ of X by connected open sets of diameter $< \varepsilon$, the Lebesgue number δ of this cover satisfies the statement, since if $d(x, y) < \delta$, $\{x, y\}$ will be contained in some element of $\mathcal{U}$. $\square$

Theorem 5.9. *A Peano continuum X is uniformly locally arc connected, that is, for each $\varepsilon > 0$ there is some δ such that if $d(x, y) < \delta$, then there is an arc joining x and y of diameter $< \varepsilon$.*

Proof. By the preceding lemma, given $\varepsilon > 0$ there will be some δ such that if $d(x, y) < \delta$, x, y lie in a connected open subspace U of diameter $< \varepsilon$. Since U is open and connected in X , by 5.6 there will be an arc between x and y in U , which will therefore have diameter $< \varepsilon$. $\square$

Theorem 5.10 (Hahn-Mazurkiewicz). *A T_2 space X is a continuous image of $\mathbf{I}$ if and only if it is a Peano continuum.*

Proof. We already proved the forward implication at the beginning of the section.

Let X be a Peano continuum. Then by the Hausdorff-Alexandroff Theorem 1.12 there is a continuous surjective function $g : \mathcal{C} \rightarrow X$. Recall that the Cantor set is obtained by removing countably many intervals from $\mathbf{I}$. We will call these intervals I_n : $I_1 = (\frac{1}{3}, \frac{2}{3})$, $I_2 = (\frac{1}{9}, \frac{2}{9})$, $I_3 = (\frac{7}{9}, \frac{8}{9})$, $I_4 = (\frac{1}{27}, \frac{2}{27})$, $\dots$

We will define our continuous function $f : \mathbf{I} \rightarrow X$, which will be an extension of g (and therefore surjective). We have to define f on $I_n = (p_n, q_n)$ for every n , and we will define it as an arc between $g(p_n)$ and $g(q_n)$. For f to be continuous, we will have to ensure that if the interval (p_n, q_n) is sufficiently small, the arc from $g(p_n)$ to $g(q_n)$ will have as small a diameter as we wish. We will be able to achieve this thanks to Lemma 5.8 and the uniform continuity of g , as we see in the following paragraph.

For each $\varepsilon > 0$, there is some $\delta > 0$ such that if $d(p, q) < \delta$, then by Lemma 5.8 any 2 points of X at distance $< \delta$ are joined by an arc of diameter $< \varepsilon$. Hence, since g is uniformly continuous, there will be some $\gamma > 0$ such that if two points of the Cantor set, x, y , are at distance $< \gamma$, then there is an arc between $g(x)$ and $g(y)$ of diameter $< \varepsilon$.

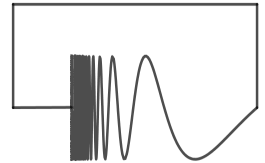
Using this, we define f on I_n as an arc between $g(p_n)$ and $g(q_n)$, so that for each $\varepsilon > 0$ there is some $\delta > 0$ such that if $q_n - p_n < \delta$, then the arc $f([p_n, q_n])$ has diameter $< \varepsilon$. That is, intuitively, as $d(p_n, q_n)$ tends to 0, the diameter of the arc $f([p_n, q_n])$ tends to 0. It is clear that f is continuous at the interior points of the intervals I_n , so let us show that it is continuous at the remaining points, which are those of $\mathcal{C}$.

Given $x \in \mathcal{C} - \{1\}$, let us show that f is continuous from the right at x (the left-hand case is symmetric). If $x = p_n$ for some n , it is obvious that f is continuous from the right at x , since it is continuous on $[p_n, q_n]$. Otherwise, let $\varepsilon > 0$. There are only finitely many intervals (p_n, q_n) such that $f([p_n, q_n])$ has diameter $> \frac{\varepsilon}{2}$. We choose $\delta > 0$ such that $(x, x + \delta) \subseteq \mathbf{I}$ does not intersect any of those intervals and such that if $y \in \mathcal{C}$ and $|x - y| < \delta$, $d(g(x), g(y)) < \frac{\varepsilon}{2}$. Now, given $y \in (x, x + \delta)$, if $y \in \mathcal{C}$ we have $d(g(x), g(y)) < \frac{\varepsilon}{2}$, and otherwise y belongs to (p_n, q_n) , with $x < p_n < y$. Thus $d(g(x), g(y)) \leq d(g(x), g(p_n)) + d(g(p_n), g(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. Hence $f((x, x + \delta)) \subseteq B(g(x), \varepsilon)$. $\square$

5.2 Continuous images of $\mathbb{R}$. Generalized Peano continua

Now that we have characterized the T_2 continuous images of $\mathbf{I}$, we could consider characterizing those of $\mathbb{R}$. The continuous images of $\mathbb{R}$ are path connected and separable, and that is where our good news ends.

A T_2 image of $\mathbb{R}$ need not be metrizable: to see this it suffices to consider the quotient of $\mathbb{R}$ obtained by identifying all the points of $\mathbb{Z}$. In this quotient, the point corresponding to $\mathbb{Z}$ has no countable neighborhood base, so the space is not metrizable. The image need not be locally connected or locally compact either, as shown by the version of the Warsaw circle in the figure, consisting of the graph of $f(x) = \sin(\frac{1}{x})$ on $(0, \frac{1}{\pi})$ and 4 closed line segments.



On the other hand, since $[0, 1]$ is a continuous image of $\mathbb{R}$, all Peano continua will be images of $\mathbb{R}$. Since the images of $\mathbb{R}$ need not be compact, we can obtain a somewhat broader class than the Peano spaces by replacing compactness with local compactness:

Definition 5.11. We say that a space X is a generalized Peano continuum if it is metrizable, connected, locally connected and locally compact.

The main result of this section will be that all generalized Peano continua are continuous images of $\mathbb{R}$. I found the concept of generalized Peano continua in [ACQ98]. According to that article, Mazurkiewicz proved this result in [Ma20], although the following proof is original. It will be analogous to the proof in the preceding section that Peano continua are continuous images of $\mathbf{I}$.

For a generalized Peano space to be a continuous image of $\mathbb{R}$, it must be path connected, which we prove in the following lemma:

Lemma 5.12. *Let X be a generalized Peano continuum and U a connected open set of X with $\overline{U}$ compact. Then U is arc connected.*

Proof. Since U is locally connected because X is, it will have a cover by connected open sets of diameter < 1 whose closures are contained in U . Therefore Lemma 5.4 tells us that, given two points a and b of U , there will be a simple chain $\mathfrak{C}_1$ from a to b of connected open sets in U of diameter < 1 .

From here we can continue the argument just as in the proof of 5.5 to obtain an arc from a to b in U . The only difference is that since X need not now be compact, we have had to consider an open set U with compact closure so that the sets C_k are compact. $\square$

Corollary 5.13. *Any generalized Peano continuum X is locally path connected, and therefore path connected (or arc connected).*

Proof. Any $x \in X$ has a base of connected neighborhoods with compact closure. $\square$

A condition that any continuous image of the real line will have to satisfy is being σ -compact, that is, being a countable union of compact subspaces. This is because if $f : \mathbb{R} \rightarrow X$ is surjective, $\{f([-n, n])\}_{n \in \mathbb{N}}$ is a countable family of compact sets covering X . That generalized Peano continua are σ -compact is a consequence of the following result (a lemma on page 460 of [Sp99]):

Lemma 5.14. *If X is a connected and locally compact metric space, then X is σ -compact.*

Proof. Since X is metric, it will be paracompact by 0.2.

Therefore we have a locally finite cover by open sets with compact closure, $\mathcal{U}$. Let $U_0 \in \mathcal{U}$. Then $\overline{U_0}$, being compact, can only intersect finitely many elements of $\mathcal{U}$, which we will call $U_1, \dots, U_{n_1}$. Similarly, $\overline{U_0} \cup \dots \cup \overline{U_{n_1}}$ only intersects finitely many elements of $\mathcal{U}$, which we call $U_{n_1+1}, \dots, U_{n_2}$. We can repeat this process infinitely many times, obtaining a sequence $(U_n)_{n \in \mathbb{N}}$, where the closure of each element is contained in the union of finitely many other elements. Therefore we can consider the union:

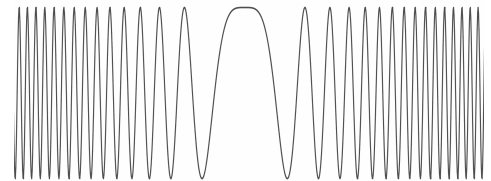
$$U = \bigcup_{n \in \mathbb{N}} U_n = \bigcup_{n \in \mathbb{N}} \overline{U_n}$$

U is clearly open, and it is closed since $\{\overline{U_n}\}_{n \in \mathbb{N}}$ is locally finite. Therefore we have $U = X$, and by the definition of U , X is a countable union of compact sets. $\square$

Corollary 5.15. *Generalized Peano continua are σ -compact (and therefore separable).* $\square$

We now proceed to give a proof that generalized Peano continua are continuous images of $\mathbb{R}$, similar in spirit to the proof that every Peano continuum is a continuous image of $[0, 1]$. First we will cover our space with the images of countably many copies of the Cantor set, which can be placed in the real line without difficulty. Then we will find a way to extend the function continuously to the whole real line.

Here we encounter a problem: unlike Peano continua, generalized Peano continua need not be uniformly locally path connected (5.8), as demonstrated by the graph of the function $\cos(x^2)$ in the figure, which is a generalized Peano continuum but does not have that property. Fortunately, we can find a weaker property that still allows us to extend the function:



Lemma 5.16. *If X is a generalized Peano continuum and C is a compact subspace of X , C is uniformly locally arc connected with respect to X , that is, for each $\varepsilon > 0$ there is some δ such that if $x, y \in C$ and $d(x, y) < \delta$, then there is an arc in X joining x and y of diameter $< \varepsilon$.*

Proof. Let $\varepsilon > 0$. Since each point of X has a base of connected neighborhoods with compact closure, by 5.12 there will be a cover of X by arc connected open sets of diameter $< \varepsilon$, $\mathcal{B}$. Writing $\mathcal{B}_C = \{B \cap C; B \in \mathcal{B}\}$, $\mathcal{B}_C$ is an open cover of C . Therefore its Lebesgue number δ satisfies the statement, since if $x, y \in C$ satisfy $d(x, y) < \delta$, then x and y belong to the same element of $\mathcal{B}$, and there is an arc in X between x and y of diameter $< \varepsilon$. $\square$

We now state the final theorem, whose proof will be very similar to that of 5.10.

Theorem 5.17. *Let X be a generalized Peano continuum. Then there is a continuous surjective function $f : \mathbb{R} \rightarrow X$.*

Proof. Let $\mathcal{C}$ be the Cantor set, let $\mathcal{C} + x = \{x + y; y \in \mathcal{C}\}$, and consider the following subset of $\mathbb{R}$ consisting of infinitely many Cantor sets with gaps between them:

$$D = \bigcup_{n \in \mathbb{Z}} \mathcal{C} + 2n$$

Then we can find a continuous surjective function $g : D \rightarrow X$. Indeed, Lemma 5.14 tells us that X is a countable union of compact sets $\{C_n\}_{n \in \mathbb{Z}}$, and by the Hausdorff-Alexandroff Theorem 1.12 there is a continuous surjective function $g_n : \mathcal{C} + n \rightarrow C_n$, and we define our g so that $g|_{\mathcal{C}+n} = g_n$.

We will define our continuous function $f : \mathbb{R} \rightarrow X$, which will be an extension of g (and therefore surjective). Again, we only have to define f on a sequence of intervals $I_n = (p_n, q_n)$, which are the gaps of D in the real line. On the interval I_n we will define f as an arc from $g(p_n)$ to $g(q_n)$. For f to be continuous, just as in the proof of 5.10, we will have to impose a condition on the diameters of $f(I_n)$:

Given two points $a, b \in X$, we define $\varrho(a, b)$ as the infimum of the diameters of the arcs from a to b . We then define f on I_n as an arc from $g(p_n)$ to $g(q_n)$ with diameter $< 2\varrho(p_n, q_n)$. The continuity of f is obvious at the interior points of the intervals I_n , so it remains to check that f is continuous at the points of D .

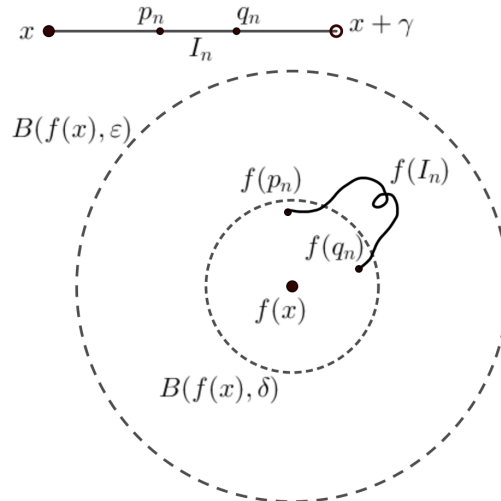


Figure 5.2: The diameter of $f(I_n)$ is $< \frac{\varepsilon}{2}$, hence $f(I_n) \subseteq B(f(x), \delta + \frac{\varepsilon}{2}) \subseteq B(f(x), \varepsilon)$.

Given $x \in D$, let us show that f is continuous from the right at x (continuity from the left is symmetric). If x is the start of an interval I_n , $x = p_n$, the result is obvious since f is continuous on $[p_n, q_n]$. Otherwise, there will be infinitely many intervals I_n approaching x from the right. This case will take a little more work; Figure 5.2 explains it geometrically.

Let $\varepsilon > 0$ be such that $\overline{B(f(x), \varepsilon)}$ is compact. Then, by Lemma 5.16, there will be some $\delta > 0$ such that if $a, b \in \overline{B(f(x), \varepsilon)}$ and $d(a, b) < 2\delta$, then there are arcs from a to b of length $< \frac{\varepsilon}{4}$. In particular, if we have endpoints (p_n, q_n) such that $f(p_n)$ and $f(q_n)$ lie in $B(f(x), \delta)$, then the diameter of $f(I_n)$ will be $< \frac{\varepsilon}{2}$. Obviously we may assume $\delta < \frac{\varepsilon}{2}$. Moreover, by continuity of g , we can choose γ so that if an interval I_n intersects $(x, x + \gamma)$, then the images of its endpoints, $f(p_n)$ and $f(q_n)$, lie in $B(f(x), \delta)$. We will be done if we show that any $y \in (x, x + \gamma)$ satisfies $d(f(x), f(y)) < \varepsilon$.

• If $y \in D$ then, since $d(x, y) < \gamma$, by continuity of g we have $d(f(x), f(y)) \leq \delta < \varepsilon$.

• Otherwise, y belongs to an interval $I_n = (p_n, q_n)$, with $f(p_n)$ and $f(q_n)$ in $B(f(x), \delta)$. Therefore $f(I_n)$ has diameter $< \frac{\varepsilon}{2}$, hence $d(f(x), f(y)) \leq d(f(x), f(p_n)) + d(f(p_n), f(y)) < \delta + \frac{\varepsilon}{2} < \varepsilon$. $\square$

5.3 A characterization of $\mathbb{R}$

This section is also original. Using the results of the previous section, we can prove a characterization of $\mathbb{R}$. To do this, we use a new concept:

Definition 5.18. We say that x is a strong cut point of X if $X - \{x\}$ has 2 connected components.

In $\mathbb{R}$, all points are strong cut points. In fact, the main theorem of this section states that this characterizes $\mathbb{R}$ among generalized Peano continua:

Theorem 5.19. *If X is a generalized Peano continuum and all its points are strong cut points, then X is homeomorphic to $\mathbb{R}$.*

This will follow from the lemmas we develop below, which are quite geometric in nature. The purpose of the following lemma is to rule out the possibility that our spaces contain subspaces homeomorphic to circles.

Lemma 5.20. *If X is connected, every point of X is a strong cut point and X contains an uncountable connected subspace A without cut points, then X is not separable.*

Proof. For each point a of A , $X - \{a\}$ has two connected components. One of them, which we call B_a , contains $A - \{a\}$ because $A - \{a\}$ is connected. We call the other one C_a . Let us show that if a, b are distinct points of A , then C_a and C_b are disjoint. This holds because if they shared a point c , then $C_a \cup C_b$ would be connected. Therefore $C_a \cup C_b \cup \{b\}$ would be connected, since b is in the boundary of C_b . This is impossible, since $C_a \cup C_b \cup \{b\}$ is a subset of $X - \{a\}$ that contains points of B_a and of C_a .

Therefore $\{C_a\}_{a \in A}$ is an uncountable set of disjoint open sets, hence X is not separable. $\square$

Lemma 5.21. *If X is connected, metric and separable and every point of X is a strong cut point, there cannot be two distinct arcs between two points. That is, given $a, b \in X$ and two arcs $f, g : [0, 1] \rightarrow X$ from a to b , we have $f([0, 1]) = g([0, 1])$.*

Proof. Suppose otherwise that there is some c with $f(c) \notin g([0, 1])$. Now let (a, b) be the maximal interval around c such that $f((a, b))$ does not intersect $g([0, 1])$. Then $f(a)$ and $f(b)$ are distinct because f is injective, and they belong to $g([0, 1])$, so there are a' and b' with $f(a) = g(a')$ and $f(b) = g(b')$. Then $f|_{[a, b]}$ and $g|_{[a', b']}$ (or $g|_{[b', a']}$, if $a' > b'$) are two arcs from $f(a)$ to $f(b)$ that meet only at their endpoints. Therefore, the union $f([a, b]) \cup g([a', b'])$ (or $g([b', a'])$) is homeomorphic to a circle, that is, an uncountable connected subspace of X without cut points. Since this contradicts the previous lemma, we are done. $\square$

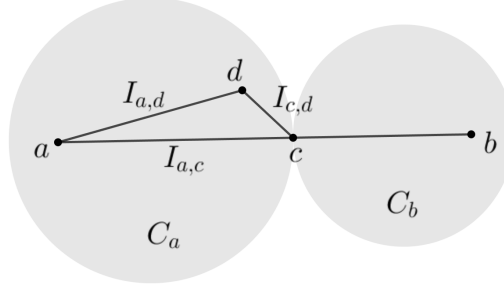
Given three points a_1, a_2, a_3 in a space X and two arcs $f_{1,2}, f_{2,3} : [0, 1] \rightarrow X$ from a_1 to a_2 and from a_2 to a_3 respectively, we can obtain an arc from a_1 to a_3 contained in the union of the previous ones. This is obvious if a_1 is in the image of $f_{2,3}$ or a_3 is in the image of $f_{1,2}$. Otherwise, let k be the maximum such that $f_{1,2}([0, k])$ does not intersect the image of $f_{2,3}$, and let k' be such that $f_{1,2}(k) = f_{2,3}(k')$. Then the path composition $f_{1,2}|_{[0, k]} \bullet f_{2,3}|_{[k', 1]}$ is an arc from a_1 to a_3 . We can therefore give the following definition:

Definition 5.22. Given two arcs $f, g : [0, 1] \rightarrow X$ with images I_f and I_g , with $f(1) = g(0)$, we denote by $I_f \wedge I_g$ the image of an arc from $f(0)$ to $g(1)$, such that $I_f \wedge I_g \subseteq I_f \cup I_g$. To avoid ambiguity, we can take the arc $I_f \wedge I_g$ constructed as in the previous paragraph.

Lemma 5.23. *Every generalized Peano continuum all of whose points are strong cut points is locally 1-Euclidean (that is, each of its points has a neighborhood homeomorphic to $\mathbb{R}$).*

Proof. Let X be a generalized Peano continuum, let $c \in X$ be any point of it that separates it into two open sets, C_a and C_b , and let $a \in C_a$ and $b \in C_b$ be two points. Since X is arc connected, there is an arc from a to b , with image I_{ab} . Since I_{ab} is connected, C_a and C_b cannot separate it, so it passes through c . It then suffices to show that $I_{ab} - \{a, b\}$ contains a neighborhood of c , since $I_{ab} - \{a, b\}$ is homeomorphic to $\mathbb{R}$.

Suppose this is not the case. Then, by Lemma 5.16, there is a point d in $X - I_{ab}$ close to c such that there is an arc from c to d , I_{cd} , of diameter $< \min(d(a, c), d(b, c))$. In particular, $a, b \notin I_{c,d}$. We assume $d \in C_a$ (the case $d \in C_b$ is symmetric). Then, since C_a is a generalized Peano continuum because it is open and connected in X , there is an arc $I_{a,d}$ from a to d in C_a .



We thus have three arcs, $I_{a,d}$, $I_{a,c}$ (part of the arc $I_{a,b}$) and $I_{c,d}$, joining the three points a, c, d pairwise, and such that $I_{a,d}$ does not pass through c , $I_{a,c}$ does not pass through d and $I_{c,d}$ does not pass through a . Let us show that this situation cannot occur. The arc $I_{a,c} \wedge I_{c,d}$ is an arc from a to d , so by 5.21, $I_{a,d} = I_{a,c} \wedge I_{c,d}$, hence $I_{a,d} \subseteq I_{a,c} \cup I_{c,d}$. Since the arc $I_{a,d}$ is connected and therefore has no partition into two nonempty closed subsets, $I_{a,d} \cap I_{a,c}$ and $I_{a,d} \cap I_{c,d}$ are not disjoint, so there is a point P in $I_{a,d} \cap I_{a,c} \cap I_{c,d}$. Therefore P cannot be c, d or a . P divides X into two connected components, X_1 and X_2 . One of them contains two of the points a, c, d ; let us say, for example, that a, c are in X_1 . Then, since X_1 is a generalized Peano continuum, there is an arc from a to c in X_1 . This arc does not contain P , so its image is different from $I_{a,c}$, which contradicts 5.21. $\square$

Proof of Theorem 5.19. X will be locally 1-Euclidean, connected, separable (by 5.15) and T_2 , so by the classification theorem for 1-manifolds 0.3, it is homeomorphic either to $\mathbb{S}^1$ or to $\mathbb{R}$. Since it has cut points, it will be homeomorphic to $\mathbb{R}$. $\square$

We have thus proved that any metric, connected, locally connected and locally compact space such that all its points are strong cut points is homeomorphic to $\mathbb{R}$. In the literature there are characterizations of $\mathbb{R}$ under apparently much weaker conditions; for example, [FK70] proves that every separable, regular, connected, locally connected space X such that all its points are strong cut points is homeomorphic to $\mathbb{R}$. However, to omit the local compactness condition, one must undertake a much more detailed study of cut points, which cannot be included here due to space constraints.

Appendix A

A stronger form of homogeneity

The methods we have used to prove the countable dense homogeneity of $\mathcal{C}, \mathbb{I}$ and $\mathbb{R}$ can be refined to obtain a property stronger than countable dense homogeneity. As a consequence, we will obtain a similar homogeneity property for $\mathbb{Q}$ (even though $\mathbb{Q}$ is not countable dense homogeneous). The content of this appendix is original, as far as I know. The property we will consider is the following:

Definition A.1. We say that a separable space X is ω -countable dense homogeneous if, given a sequence A_n of pairwise disjoint countable dense sets in X , and another sequence B_n of the same form, there is a homeomorphism $f : X \rightarrow X$ with $f(A_n) = B_n$ for every n .

To prove that $\mathcal{C}$ has this property, we will consider the following space homeomorphic to $\mathcal{C}$:

$$\mathcal{C}' = \prod_{n=1}^{\infty} K_n, \text{ with } K_n = \{1, \dots, n\}$$

that is, $\mathcal{C}'$ is the space of sequences $s = (s_n)_{n \in \mathbb{N}}$, with $s_n \in \{1, \dots, n\}$, with the product topology.

To give a base of $\mathcal{C}'$, we consider the set Γ of finite sequences $\xi = (\xi_1, \dots, \xi_n)$ with $1 \leq \xi_i \leq i \forall i$. Then a base of open sets of $\mathcal{C}'$ consists, for each $\xi \in \Gamma$, of the open set B_ξ of all sequences in $\mathcal{C}'$ that begin with ξ .

Now, for $n \geq 1$, we can define the following subsets of $\mathcal{C}'$:

$$D_n = \{s \in \mathcal{C}'; (s_k) \text{ is eventually } n\}$$

The D_n are countable dense subspaces of $\mathcal{C}'$, and they are certainly pairwise disjoint. The theorem we aim to prove is the following:

Theorem A.2. Let $(E_n)_{n \in \mathbb{N}}$ be a sequence of pairwise disjoint countable dense subspaces of $\mathcal{C}$. Then there is a homeomorphism $f : \mathcal{C}' \rightarrow \mathcal{C}$ with $f(D_n) = E_n$ for every n .

Proof. Let $(A_n)_{n \in \mathbb{N}}$ be a base of clopen sets of $\mathcal{C}$. For each E_n we give an enumeration $(x_{n,k})_{k \in \mathbb{N}}$. Now, for each open subset Y of $\mathcal{C}$, we define $e_{Y,n}$ as the first element of E_n that belongs to Y (according to the enumeration of E_n we have given).

Similarly to the proof of 1.6, we will define a set I_ξ for each $\xi \in \Gamma$. The aim will be to construct the function f in the statement, so that $f(B_\xi) = I_\xi$ for every ξ . The I_ξ will satisfy these properties:

- (1) For every $\xi \in \Gamma$, I_ξ is a nonempty clopen subset of $\mathcal{C}$ (and therefore is homeomorphic to $\mathcal{C}$).
- (2) If ξ has length n , then the $I_{\xi \wedge (i)}$ with $i = 1, \dots, n+1$ form a partition of I_ξ . Moreover, for i between 1 and n , $I_{\xi \wedge (i)}$ contains the point $e_{I_\xi, i}$.
- (3) If ξ has length n , I_ξ is either contained in A_n or disjoint from A_n .

We will define the I_ξ recursively; in the base case we define $I_{(1)} = \mathcal{C}$. Now, suppose we have defined a nonempty clopen I_ξ for some ξ of length n , with $n \geq 1$, and we will define $I_{\xi \wedge (i)}$ for $i = 1, \dots, n+1$.

For i between 1 and n we define $I_{\xi \wedge (i)}$ as a clopen neighborhood of $e_{I_{\xi}, i}$ contained in I_{ξ} , so that the $I_{\xi \wedge (i)}$ from 1 to n are pairwise disjoint, each of them is contained in A_n or disjoint from A_n , and they cover neither $I_{\xi} \cap A_n$ nor $I_{\xi} \setminus A_n$. Now, if $e_{I_{\xi}, 1}$ is in A_n , we change $I_{\xi \wedge (1)}$ to $(I_{\xi} \cap A_n) \setminus \bigcup_{i=2}^n I_{\xi \wedge (i)}$ (it is still a neighborhood of $e_{I_{\xi}, 1}$), and define $I_{\xi \wedge (n+1)}$ as $I_{\xi} \setminus \bigcup_{i=1}^n I_{\xi \wedge (i)}$. If, on the other hand, $e_{I_{\xi}, 1}$ is not in A_n , we change $I_{\xi \wedge (1)}$ to $(I_{\xi} \setminus A_n) \setminus \bigcup_{i=2}^n I_{\xi \wedge (i)}$, and define $I_{\xi \wedge (n+1)}$ as $I_{\xi} \setminus \bigcup_{i=1}^n I_{\xi \wedge (i)}$.

Then, by the way we have constructed them, it is clear that the I_{ξ} satisfy properties (1), (2) and (3). Moreover, these other properties will hold:

- (4) For every n , $\{I_{\xi}; \xi \in \Gamma \text{ of length } n\}$ is a partition of $\mathcal{C}$ into $n!$ clopen sets.
- (5) For each $(a_n) \in \mathcal{C}'$, $I_{a|_n}$ is a decreasing sequence of closed sets such that $\bigcap_{n \in \mathbb{N}} I_{a|_n}$ has 1 point.

Indeed, (4) follows directly by induction and by (2), while given $(a_n) \in \mathcal{C}'$, by construction $I_{a|_{n+1}}$ is contained in $I_{a|_n}$ for every n , so the $I_{a|_n}$ form a decreasing sequence. Their intersection is nonempty because this is a chain of closed sets in $\mathcal{C}$ (which is compact) with the finite intersection property. Now suppose that $x \neq y$ both belong to $\bigcap_{n \in \mathbb{N}} I_{a|_n}$. But there will be some N such that $x \in A_N, y \notin A_N$. Thus x, y cannot both belong to $I_{a|_N}$ by (3), a contradiction.

We can now define our f :

$$\begin{aligned} f: \quad \mathcal{C}' &\rightarrow \mathcal{C}; \\ (a_n) &\mapsto y \quad , \text{ where } \{y\} = \bigcap_{n=1}^{\infty} I_{a|_n}. \end{aligned}$$

f is well defined by (5). We will prove the statement in parts:

- a) $f(B_{\xi}) \subseteq I_{\xi}$ for every ξ . This is obvious from the definition of f .
- b) f is injective and continuous. If $a = (a_n)$ and $b = (b_n)$ are two distinct sequences in $\mathcal{C}'$ and the first term at which they differ is $a_N \neq b_N$, then $I_{a|_N}$ and $I_{b|_N}$ are disjoint by (4), so $f(a) \neq f(b)$. To show that f is continuous, let $a = (a_n) \in \mathcal{C}'$ and let A_k be a neighborhood of $f(a)$ from the base of $\mathcal{C}$. Then by a), $f(a) \in f(B_{a|_k}) \subseteq I_{a|_k}$, so since $f(a) \in A_k \cap I_{a|_k}$, by (3) we have $I_{a|_k} \subseteq A_k$. Thus the neighborhood $B_{a|_k}$ of a satisfies $f(B_{a|_k}) \subseteq A_k$.
- c) f is a homeomorphism. Since $\mathcal{C}'$ is compact and $\mathcal{C}$ is T_2 , it suffices to show that f is surjective. Given $x \in \mathcal{C}$, we can recursively construct a sequence (a_n) such that $x \in I_{a|_n}$ for every n . Thus $x = f((a_n))$, and f is surjective.
- d) $f(D_n) = E_n$ for every n . We will check this for a fixed n . Let us show that $f(D_n) \supseteq E_n$. Given $x \in E_n$, let $a = (a_i)$ be the sequence such that $x = f(a) = \bigcap_{i=1}^{\infty} I_{a|_i}$. Since the intersection of the $I_{a|_i}$ is a single point, there will be some $N > n + 1$ such that $I_{a|_N}$ contains no points of E_n preceding x in the enumeration of E_n . That is, $x = e_{I_{a|_N}, n}$. Therefore, using (2), we have $x \in I_{a|_{N \wedge (n)}}$, so since we also have $x \in I_{a|_{N+1}}$, we have $a_{N+1} = n$. Moreover, x is the first point of E_n in $I_{a|_{N+1}}$. We thus continue with an inductive argument, showing that $a_i = n$ and $x \in I_{a|_i}$ for every $i > N$, so that a_i is n from some i onwards, hence $a \in D_n$.

Now let us show that $f(D_n) \subseteq E_n$. Given $a = (a_i) \in D_n$, there is some $N > n + 1$ such that $a_i = n$ for $i \geq N$. Then, I write $x = e_{I_{a|_N}, n} \in E_n$, and it suffices to show that $x = f(a)$. This holds since, as $x = e_{I_{a|_N}, n}$ and $a_{N+1} = n$, by (2) we will have $x \in I_{a|_{N+1}}$, and we also have $x = e_{I_{a|_{N+1}}, n}$, and by induction we see that $x \in I_{a|_i}$ for every $i \geq N$, so $x = f(a)$.

□

Corollary A.3. $\mathcal{C}$ is ω -countable dense homogeneous.

□

Corollary A.4. $\mathbb{I}$ is ω -countable dense homogeneous.

Proof. By 2.12, $\mathbb{I}$ is homeomorphic to the complement of any countable dense subset of $\mathcal{C}$ (since this complement will be completely metrizable because it is G_{δ} in $\mathcal{C}$ and nowhere locally compact by 2.2). I denote by I a subset of $\mathcal{C}$ such that $\mathcal{C} \setminus I$ is countable and dense in $\mathcal{C}$, and let $(A_n)_{n \geq 2}$ and $(B_n)_{n \geq 2}$ be two sequences of pairwise disjoint countable dense sets in I . Then we can consider (A_n) and (B_n) as sequences of countable

dense sets in $\mathcal{C}$, and setting $A_1 = B_1 = \mathcal{C} \setminus I$ and using the ω -countable dense homogeneity of $\mathcal{C}$, we have that I is ω -countable dense homogeneous. $\square$

Corollary A.5. *If A_n, B_n are sequences of disjoint countable dense sets in $\mathbb{Q}$ with $\bigcup_{n \in \mathbb{N}} A_n = \mathbb{Q} = \bigcup_{n \in \mathbb{N}} B_n$, then there is a homeomorphism $f : \mathbb{Q} \rightarrow \mathbb{Q}$ with $f(A_n) = B_n$.*

Proof. Similarly to the preceding corollary, we consider $\mathbb{Q}$ as a countable dense subset of $\mathcal{C}$ and apply the ω -countable dense homogeneity of $\mathcal{C}$. $\square$

Next we check that $\mathbb{R}$ also has this property of ω -countable dense homogeneity. The approach to proving it will be the same as for the countable dense homogeneity of $\mathbb{R}$, so we begin by proving a lemma about order. Given an ordered set X , we will say that a subset Y is dense in X if it is dense in the order topology. In particular, if X has a dense order with no greatest or least element and Y is dense in X , then there are elements of Y in (a, b) , (a, ∞) and $(-\infty, b)$ for each $a, b \in X$ with $b > a$.

Lemma A.6. *Let A, B be ordered sets that are order-isomorphic to $\mathbb{Q}$, and let A_n and B_n be sequences of pairwise disjoint dense subsets of A and B respectively, such that $\bigcup_{n=1}^{\infty} A_n = A$, $\bigcup_{n=1}^{\infty} B_n = B$. Then there is an order isomorphism $f : A \rightarrow B$ with $f(A_n) = B_n$ for every n .*

Proof. We construct the isomorphism recursively, in almost the same way as in 4.3. First, some notation: let $(a_n)_{n \in \mathbb{N}}$ be an enumeration of A and let $(b_n)_{n \in \mathbb{N}}$ be an enumeration of B . Moreover, I denote by $e(a_n)$ the unique natural number k such that $a_n \in A_k$, and by $e(b_n)$ the unique natural number k such that $b_n \in B_k$. Thus what the statement asks for is an order isomorphism $f : A \rightarrow B$ such that $e(f(a_n)) = e(a_n)$ for every n .

We will construct f recursively. We begin by defining $f(a_1)$ as b_k , where k is the smallest natural number such that $e(b_k) = e(a_1)$. And, if $f(a_1) \neq b_1$, we define $f^{-1}(b_1)$ as a_k , where k is the smallest natural number > 1 such that $e(a_k) = e(b_1)$.

Now suppose that we have already defined $f(a_1), \dots, f(a_{n-1}), f^{-1}(b_1), \dots, f^{-1}(b_{n-1})$ so that f is increasing and injective on its domain, and $e(f(a_n)) = e(a_n)$ for every a_n in its domain. We will define $f(a_n)$ and $f^{-1}(b_n)$. If a_n was not already in the domain of f , we define $f(a_n)$ as b_k , where k is the smallest index such that:

- We have not yet defined $f^{-1}(b_k)$.
- We have $e(b_k) = e(a_n)$.
- Defining $f(a_n) = b_k$ leaves f increasing.

Such a k will exist because $B_{e(a_n)}$ is dense in B , so we can find elements of $B_{e(a_n)}$ in any interval of B . Now, similarly, we can define $f^{-1}(b_n)$ (if b_n was not already in the image) so that f remains increasing and injective and $e(f^{-1}(b_n)) = e(b_n)$ (this time we use the fact that $A_{e(b_n)}$ is dense in A).

After defining $f(a_n)$ and $f^{-1}(b_n)$ for each n by this process, we obtain an injective, increasing f with $e(f(a_n)) = e(a_n)$ for every n , and moreover f is surjective because we have defined $f^{-1}(b_n)$ for every n . Therefore we now have the desired order isomorphism. $\square$

Theorem A.7. *The real line $\mathbb{R}$ is ω -countable dense homogeneous.*

Proof. Let A_n, B_n be sequences of pairwise disjoint countable dense sets in $\mathbb{R}$. We consider $A = \bigcup_{n=1}^{\infty} A_n$ and $B = \bigcup_{n=1}^{\infty} B_n$. Then A and B are also countable dense subsets of $\mathbb{R}$, so by 4.3 they are order-isomorphic to $\mathbb{Q}$. By the preceding lemma, we have an order isomorphism $g : A \rightarrow B$ with $g(A_n) = B_n$ for every n . Moreover, in the proof of 4.4 we showed that every order isomorphism between two countable dense subsets of $\mathbb{R}$ can be extended to a homeomorphism from $\mathbb{R}$ to $\mathbb{R}$. Thus, writing f for this extension of g , we now have a homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(A_n) = B_n$ for every n . $\square$

Appendix B

$2^{\aleph_0}$ distinct subspaces of $\mathbb{Q}$

This appendix can be seen as a continuation of Section 3.2. In that section we described uncountably many pairwise nonhomeomorphic subspaces of $\mathbb{Q}$. A natural question is whether there are in fact as many subspaces of $\mathbb{Q}$ distinct up to homeomorphism as there are subsets of $\mathbb{Q}$, that is, $2^{\aleph_0}$. If you assume the continuum hypothesis, we have already answered the question, but even if you do not assume it, it is not difficult to answer this question in the affirmative. To check this, we recall these definitions of subsets of $\mathbb{Q}$, for a natural number n (the first cases are shown in the figure in Section 3.2).

$$\begin{cases} M_0 &= \{0\} \\ M_{n+1} &= \{0\} \cup \bigcup_{i=1}^{\infty} \left\{ \frac{1}{2^i} + \frac{M_n}{2^{i+1}} \right\} \end{cases}$$

and add a very similar new class of sets. For $n \geq 2$:

$$N_n = M_n \setminus \left\{ \frac{1}{2^i}; i \in \mathbb{N} \right\}$$

Proposition B.1. *0 is an $n-1$ -isolated point of N_n .*

Proof. M_n , with $n \geq 2$, consists of 0 and countably many copies of M_{n-1} that are clopen in M_n . Moreover, 0 is an n -isolated point in M_n , so the points of the form $\frac{1}{2^i}$ are the $n-1$ -isolated points in M_n (since they correspond to the zeros of the copies of M_{n-1} in M_n). Therefore N_n is obtained from M_n by removing its $n-1$ -isolated points. N_n contains 0 and copies of $M_{n-1} - \{0\}$ that are clopen in M_n . Since in turn $M_{n-1} - \{0\}$ consists of clopen copies of M_{n-2} , N_n can be partitioned into 0 and countably many copies of M_{n-2} that are clopen in N_n .

Therefore, in $N_n - \{0\}$, the $n-2$ -isolated points are the zeros of the copies of M_{n-2} . Their only accumulation point in N_n is 0, so in N_n , 0 is an $n-1$ -isolated point. $\square$

Proposition B.2. *For $n \geq 2$, M_n is compact, and every point of N_n has compact neighborhoods except 0.*

Proof. We already saw in Section 3.2 that M_n is compact. However, N_n is not, since N_n does not contain the points $\frac{1}{2^i}$ but contains sequences that converge to them in $\mathbb{R}$. In particular, 0 has no compact neighborhood in N_n , although the remaining points of N_n do have compact neighborhoods, since they are contained in a copy of M_{n-2} that is clopen in N_n . $\square$

Thus we can now construct our $2^{\aleph_0}$ subsets distinct up to homeomorphism: it suffices to construct a set O_s for each sequence of zeros and ones $s = (s_n)_{n \in \mathbb{N}} \in 2^{\mathbb{N}}$. The construction is simply

$$O_s = \bigcup_{n \geq 2; s_n = 1} n + N_{n+1},$$

and by what we have seen about N_n , O_s will have an n -isolated point without compact neighborhoods if and only if $s_n = 1$, so O_s will not be homeomorphic to O_r if $s \neq r$.

To find these spaces, we have constructed for each n two distinct types of n -isolated points. In [Gi05] there is a study of countable metric spaces and the theory of types of their accumulation points.

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